



Industry 4.0 technologies moderately spurred micro-level circular economy considering cleaner production, not promoting sustainable performance

G. C. de Oliveira Neto¹ · D. da Silva² · V. D. Arns³ · H. N. P. Tucci⁴ · L. F. R. Pinto³ · M. N. Seri¹

Received: 23 June 2023 / Revised: 24 June 2024 / Accepted: 19 August 2024

© The Author(s) under exclusive licence to Iranian Society of Environmentalists (IRSEN) and Science and Research Branch, Islamic Azad University 2024

Abstract

The firms from textile sector are increasingly integrating industry 4.0 technologies alongside micro-level circular economy, considering cleaner production to enhance sustainable performance. The objective of this study is to assess the extent to which industry 4.0 technologies fosters circular economy practices at the micro level, considering cleaner production, thereby bolstering sustainable performance within large textile industry located in Brazil. A systematic literature review was developed with the aim of evaluating the relationship between Industry 4.0 Technologies, Circular Economy, Cleaner Production and Performance. To this end, the current state of the art on this subject was identified, however, no exploratory studies were identified. Survey method was employed, complemented by expert analysis to refine the research instrument. Structural equation modeling was utilized for hypothesis testing, alongside Pearson correlation analysis to examine relationships between variables. Findings indicate that the adoption of industry 4.0 technologies within large textile industry located in Brazil has moderately spurred micro-level circular economy initiatives, because a lack of established mechanisms for valorizing residue and restoring the value of products and packaging after consumption. In addition, sustainable performance was not validated because the focus was only on environmental and economic performance aimed at supplying the foreign market and moderately influenced social performance. Another result showed that the cloud computing, cybersecurity systems, and additive manufacturing did not stimulate micro-level circular economy actions and performance. On the other hand, the industry 4.0 technologies that boosted micro-level circular economy and performance were cyber physical system, autonomous robots, augmented reality, big data analytics, simulation, artificial intelligence and the internet of things.

Keywords Industry 4.0 technologies · Micro-level circular economy · Cleaner production · Sustainable performance · Textile firms

Editorial responsibility: S. Mirkia.

✉ G. C. de Oliveira Neto
geraldoproduct@gmail.com

¹ Industrial Engineering Post Graduation Program, Federal University of ABC, São Bernardo do Campo, São Paulo, SP 09606-045, Brazil

² Department of Cultural Practices of Campinas University (UNICAMP), Street. Bertrand Russell, 801, Cidade Universitária, Campinas, São Paulo 13083-865, Brazil

³ Industrial Engineering Post-Graduation Program, Universidade Nove de Julho (UNINOVE), Vergueiro Street, 235/249 – 12 Floor, Liberdade, São Paulo zip code: 01504-001, Brazil

⁴ Department of Science, Technology and Engineering, University Center of the Faculdades Metropolitanas Unidas (FMU), 917, Brigadeiro Luís Antônio Avenue, São Paulo zip code: 01317-001, Brazil

Introduction

The integration of Industry 4.0 technologies (I4.0 T) within companies has significantly bolstered economic, environmental, and social benefits, thereby fostering advancements conducive to promoting sustainability (Luthra and Mangla 2018). Industry 4.0 Technologies encompass a spectrum of cutting-edge advancements including big data analytics, autonomous robotics, simulation methodologies, embedded systems, the Internet of Things (IoT), cloud computing infrastructure, additive manufacturing processes, augmented reality interfaces, cyber-physical systems, and advanced security frameworks, all underpinned by artificial intelligence algorithms. (Rüßmann et al., 2015). Although the implementation of I4.0 T is significant, it alone does not ensure sustainable outcomes. It is crucial to emphasize the promotion of



Circular Economy and Cleaner Production (CECP) initiatives to achieve economic, environmental, and social benefits within production processes. CECP practices complement the adoption of I4.0 T, serving as a pivotal avenue for enhancing sustainable production performance. (Rajput and Singh 2019; Aranda-Uson et al. 2020). Thus, the use of CP enhances circular economy (CE) outcomes (Ghisellini et al. 2016; Hens et al. 2018; Shirvanimoghaddam et al. 2020), because CP is a strategic tool of the micro level of CE, drawing from preventive environmental actions undertaken by either individual companies or consumer moving towards enhancing processes, products, and services to bolster overall economic efficiency while mitigating harm and risks to both human welfare and the ecosystem. (Ghisellini et al. 2016; Franco 2017; Hens et al. 2018; Sousa-Zomer et al. 2018).

CP entails the ongoing implementation of strategies aimed at enhancing efficiency of processes and products by minimizing the utilization of raw materials, water, and energy, while reducing waste generation through the prevention, minimization, and recycling measures (UNIDO/UNEP, 1995). CP strategies play a vital role in water conservation, waste reduction (Gumbo et al., 2003), and emissions within industrial settings (Muredzi 2015). CP represents a proactive strategy that promotes resource recovery by optimizing the use of materials, water and energy, enhancing production efficiency, and eliminating toxic materials from manufacturing processes (de Oliveira Neto et al. 2023).

In this context, CE originates from and is associated with UNIDO/UNEP programs and national CP centers (Kalmykova et al., 2018; Fahimnia et al. 2017). CE is a burgeoning paradigm that prioritizes the sustainable utilization of natural resources. (Mcdowall et al. 2017). Thus, CE considers that production should replace the linear flow of material and energy with cyclical material flows with renewable energy and material sources. Effective CE practices contribute to the tripartite dimensions of sustainable development. (Korhonen et al. 2018). With that, CE aims at circularity (MacArthur et al. 2015). Hence, CE endeavors to optimize the circular flow of resources and energy within manufacturing processes, recognizing the scarcity of natural resources, to preserve value within the system by eliminating end-of-life waste (Ghisellini et al. 2016).

The textile sector is implementing IT4.0 while aiming to improve sustainable performance in production through practices in CECP (Shayganmehr et al. 2021). It is emphasized that the textile industry generates negative environmental impact due to excessive energy consumption and use of hazardous materials (Oliveira Neto et al. 2020, Oliveira Neto et al., 2021ab), an important aspect is that in a World Bank study found that 20% of water pollution is the responsibility of the textile industry (Periyasamy et al. 2017). Moreover, in the last two decades the textile industry has

doubled production, increasing the annual consumption of textiles from 7 to 13 kg per person and reached the threshold of 100 million tons of textile consumption, generating excessive volume of waste (Shirvanimoghaddam et al. 2020).

In this context, only exploratory studies in the textile industry on the relationship between I4.0 T, CECP and performance were identified through systematic literature review. Seven studies developed case study and simulation. Silva and Morais (2022) devised a simulation framework aimed at categorizing circular strategies pertinent to the outsourcing of textile waste management in Brazil. Kumar et al. (2022) conducted an analysis of advanced environmental management strategies, focusing on CECP principles, utilizing simulation, augmented reality, and cyber-physical systems to mitigate pollution within the supply chain originating from Pakistan. Agarwal and Singh (2022) employed simulation techniques to assess the efficacy of textile effluent treatment methods in India, resulting in enhanced water circularity and the reduction of hazardous byproducts. Shayganmehr et al. (2021) examined the significance of Industry 4.0 Technologies (I4.0 T) in facilitating the adoption of CECP principles within ethically conscious societies, evaluating the preparedness of an Iranian industrial context. Bai et al. (2020) investigated the applications and ramifications of I4.0 T on sustainable performance within China. Tsai (2018) discussed the integration of simulation and sensor technologies to realize CECP-aligned green production planning and control strategies in Taiwan. Bloomfield and Borstrock (2018) explored the hurdles associated with implementing additive manufacturing to advance CECP initiatives in the United Kingdom.

In addition, six literature review studies mentioned only qualitative evidence about the junction between I4.0 T, CECP and performance. Jin et al. (2021) reported on business models of apparel companies. Chen et al. (2020) proposed integrated strategies for adopting CECP-based green chemistry, finding that big data analytics and IoT are necessary for sustainable governance. Shirvanimoghaddam et al. (2020) elucidated the integration of additive manufacturing, IoT, cyber-physical systems, and artificial intelligence in advancing circular fashion. Jia et al. (2020) delineated four drivers, barriers, practices, and indicators pertaining to sustainable performance in the implementation of circular economy principles within the textile and apparel industry, highlighting the role of IoT and cyber-physical systems. Faria et al. (2020) presented mobile applications leveraging cloud computing and IoT to facilitate online apparel commerce within the CECP. Alcaayaga et al. (2019) expounded upon the intricate interplay among big data, IoT, and cyber-physical systems within smart circular systems in production process.

Also, only one case study was identified in Italy that contributes to the scientific knowledge about the characteristics



of circular business models in the CE biological cycle and mentioned about big data and cyber physical systems (Angelis and Feola 2020). Another study developed multiple case studies in Austria and Italy, one being conducted in the textile industry, mentioning about the adoption of circular production with the use of IoT and cyber-physical and security systems (Franco 2017). A study utilized focus groups to devise an intelligent and sustainable waste management system in Indonesia (Fatimah et al. 2020). Another study developed action research to evaluate four projects on sustainable development in Sweden, finding the need for the focus on augmented reality in the customer interaction process (Larsen 2018). Only one study has employed an exploratory survey coupled with ANOVA analysis to evaluate the extent of sustainability resilience according to enterprise magnitude within the textile sector, aimed at fostering incremental transitions from CP to CE, with references to additive manufacturing in the realm of product design and integration of autonomous systems (Oliveira Neto et al. 2022). Therefore, the methods used in these works can be seen in Fig. 1.

Thus, no research has been identified that has confirmatively assessed the relationship between I4.0 T, CECP practices and sustainable performance in large textile industry located in Brazil (LTILB). Based on this gap, three research questions are suggested: (QP1) Does the adoption of I4.0 T by LTILB stimulate CECP? If it does, (QP2) Which I4.0 T are related to CECP practices in LTILB? and (QP3) Does the implementation of I4.0 T related to CECP practices by LTILB have positive or negative effect on sustainable performance (economic, environmental, and social)? It is noteworthy that the Brazilian textile sector occupies a prominent place in the economy due to its job-generating capacity. Brazil is currently a reference in some fashion segments such as jeans wear and beach wear, being one of the five largest textile industrial parks in the world (ABIT, 2019).

This study aims to assess the potential of I4.0 T in stimulating CECP, promoting sustainable performance in LTILB. To achieve this objective, two specific objectives were

developed: (1) To evaluate the relationship between I4.0 T and CECP actions in LTILB; (2) To evaluate the sustainable performance (economic, environmental, and social) of LTILB that use I4.0 T related to CECP practices.

Literature review

The search for papers related to the scope of this study considered the following keywords: "textile" AND "circular economy" AND "cleaner production" AND "Industry 4.0" OR "Internet of things" OR "Big data" OR "Augmented Reality" OR "System Integration" OR "Additive Manufacturing" OR "Cyber physical System" OR "sensors" OR "Cybersecurity System" OR "Cloud Computer" OR "Simulation" OR "Autonomous Robots" OR "Artificial Intelligence". Eighteen articles were found, as shown in Figs. 2, 3, 4, 5 and 6. The content analysis verified the relationship between IT4.0, CECP and performance. It is emphasized that the CECP actions for the textile industry were organized into six Circular Economy groups whose anagram forms the word ReSOLVE, that is: Circular Economy Regenerate (CERE—Fig. 1), Circular Economy Share (CES—Fig. 2), Circular Economy Optimize (CEO—Fig. 3), Circular Economy Loop (CEL—Fig. 4), Circular Economy Virtualize (CEV—Fig. 5) and Circular Economy Exchange (CEE—Fig. 5) (MacArthur, 2015).

In addition, based on the content of the articles (scientific evidence only) 26 indicators representing the sustainable performance of research conducted in the textile industry were found. Sustainable performance is achieved when it has environmental, economic, and social results simultaneously (Elkington 1997; Gomes et al. 2020). Environmental performance (Fig. 7) in the textile industry aims to reduce/eliminate the volume of wastewater due to the fabric painting process and fabric waste generation, air emissions, energy consumption, hazardous materials, material use, and compliance and improvement with the environmental standards of the company. It enhances economic performance (Fig. 8) by mitigating costs related to material purchases, energy consumption, waste management, discharge, and penalties for environmental incidents (Jia et al. 2020). Social performance (Fig. 9) encompasses the textile industry's capacity to enhance stakeholder welfare, promote community health and safety, mitigate public risks, ensure occupational health and safety for employees, foster a positive organizational climate, reduce absenteeism, and stimulate job creation (Shayganmehr et al. 2021).

IT4.0, CECP and performance

Big data has been adopted in some research conducted in the textile sector with a focus on organizational decision

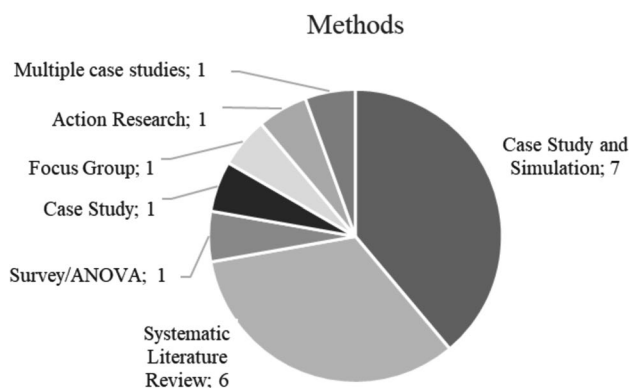


Fig. 1 Methods adopted by relevant works



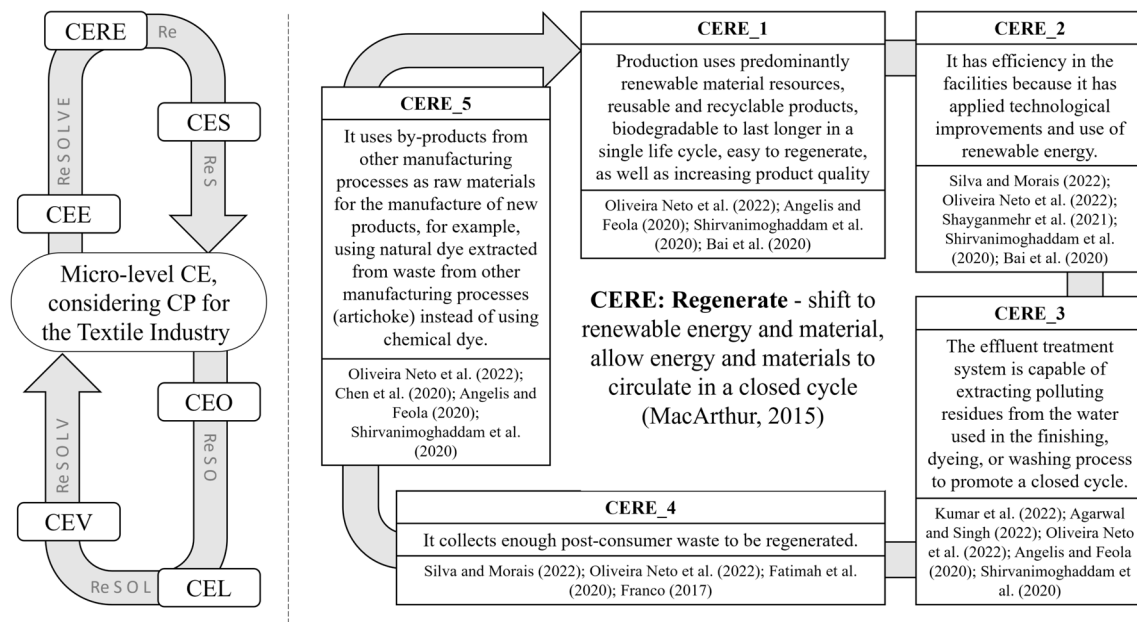


Fig. 2 Six groups of the ReSOLVE and Regenerate step (CERE)

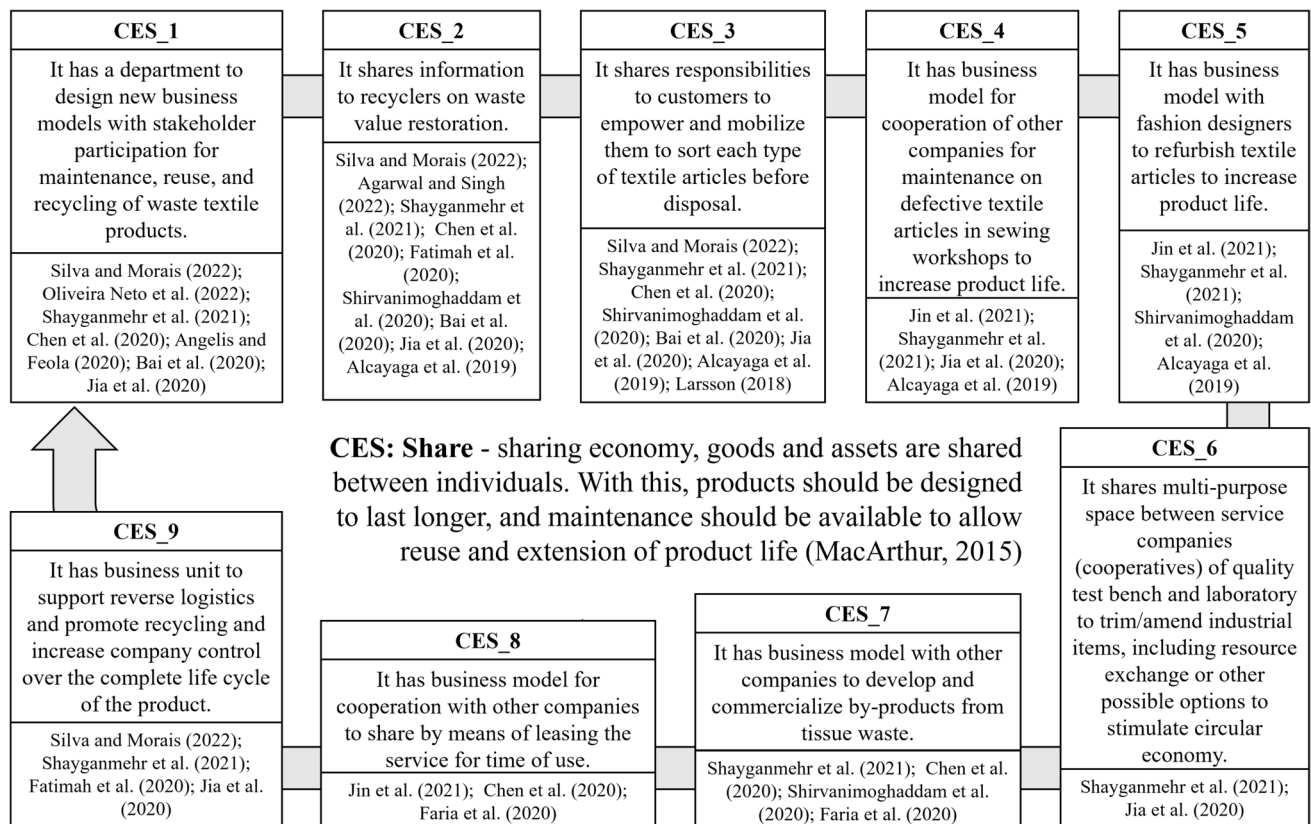


Fig. 3 Share step (CES)



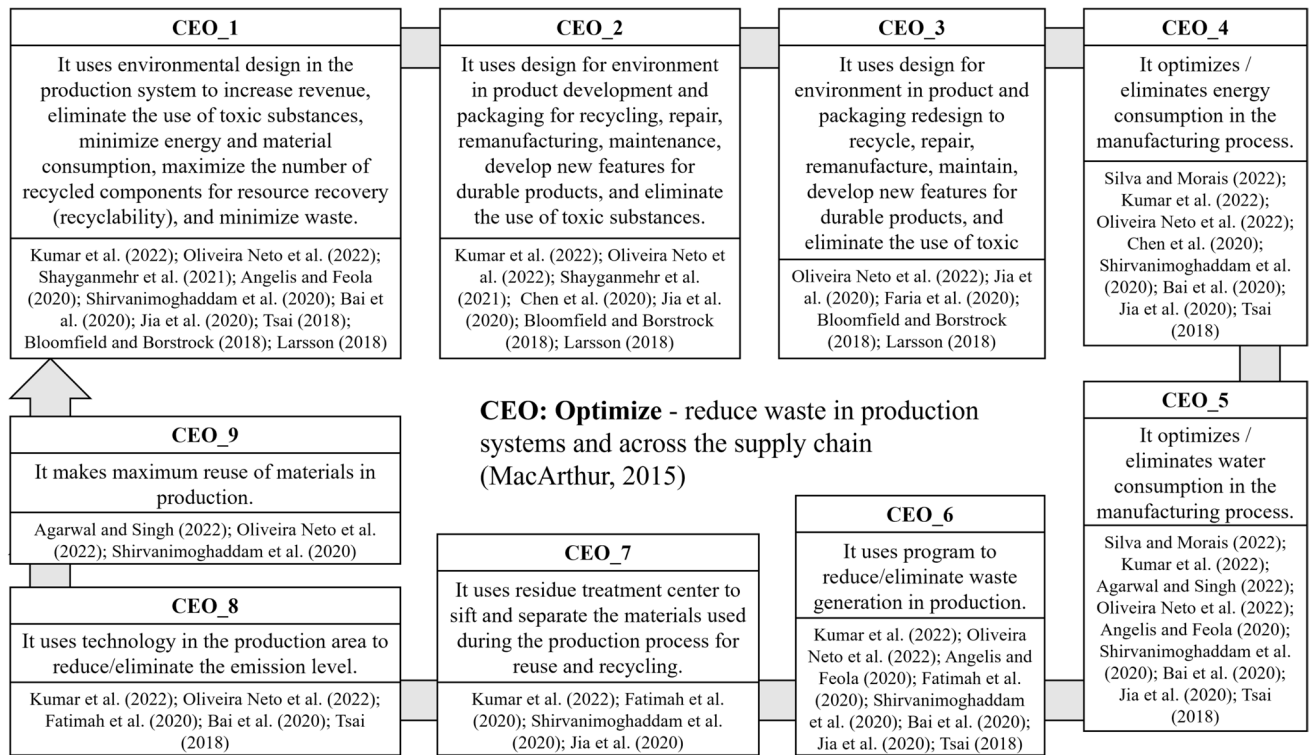


Fig. 4 Optimize step (CEO)

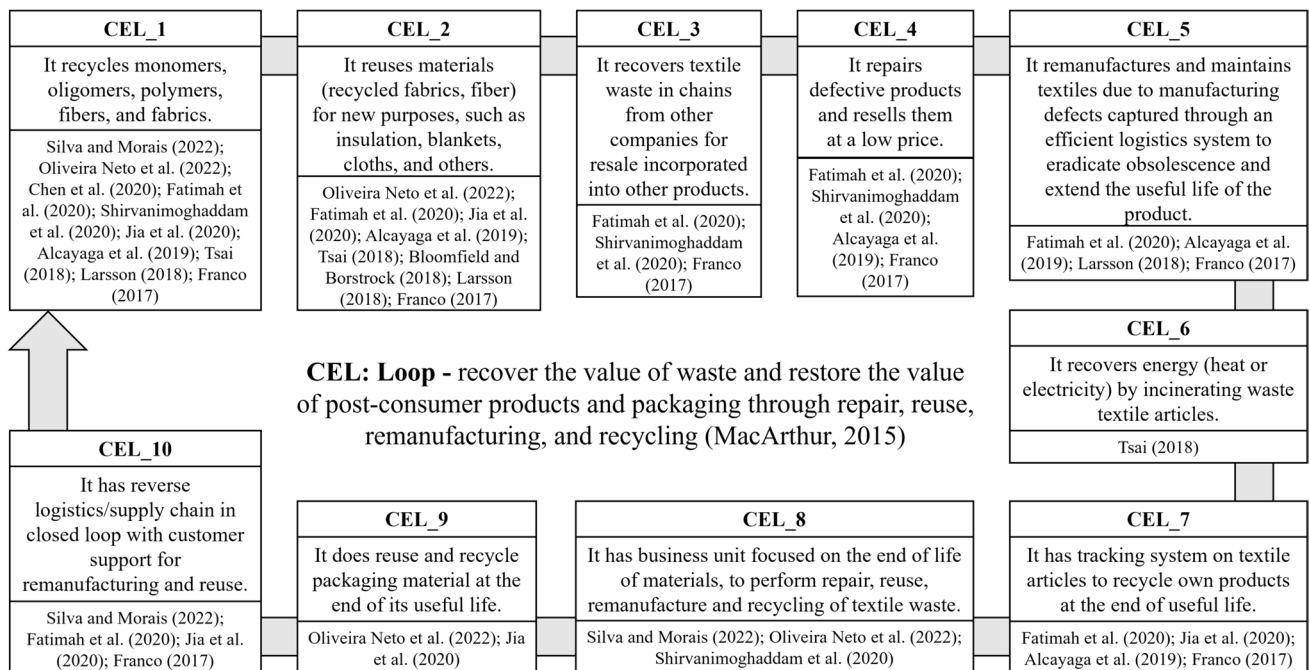


Fig. 5 Loop step (CEL)



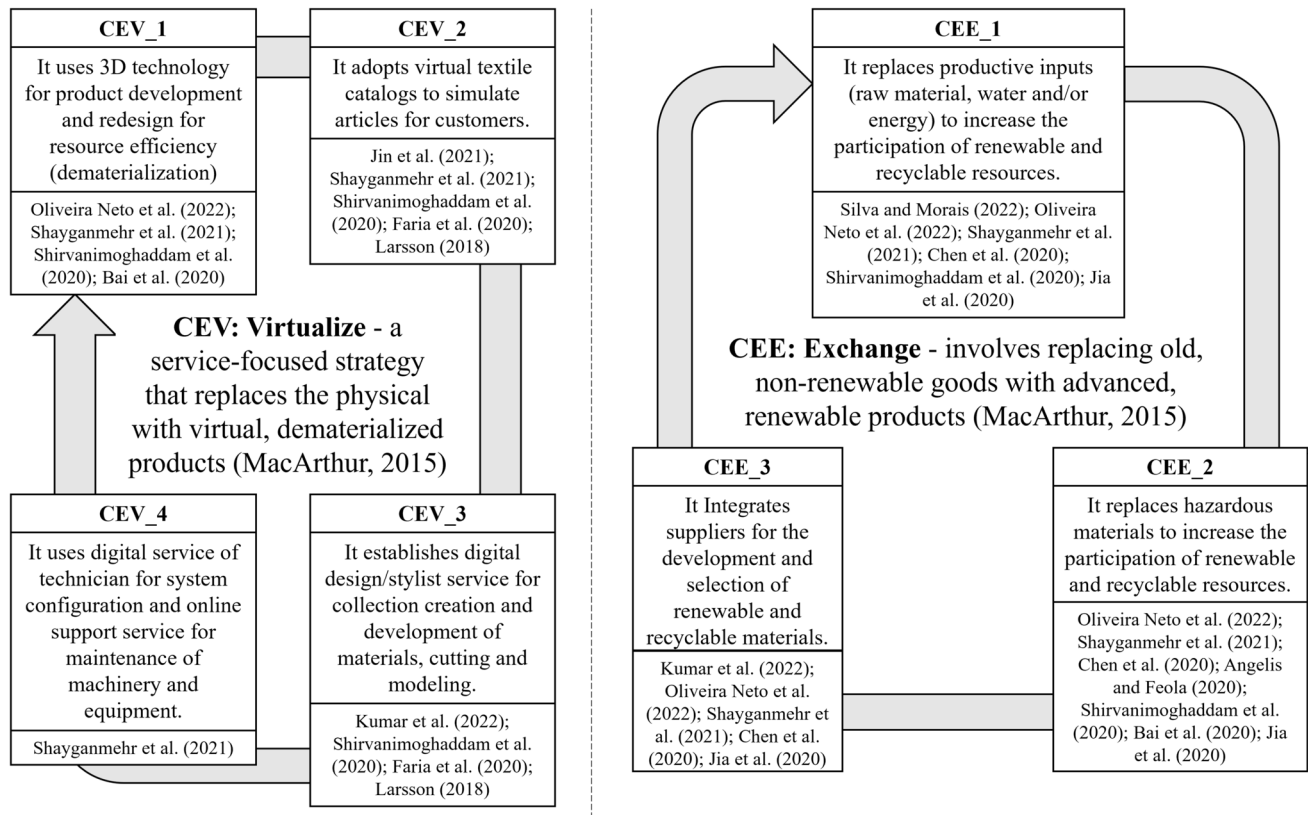


Fig. 6 Virtualize step (CEV) and Exchange step (CEE)

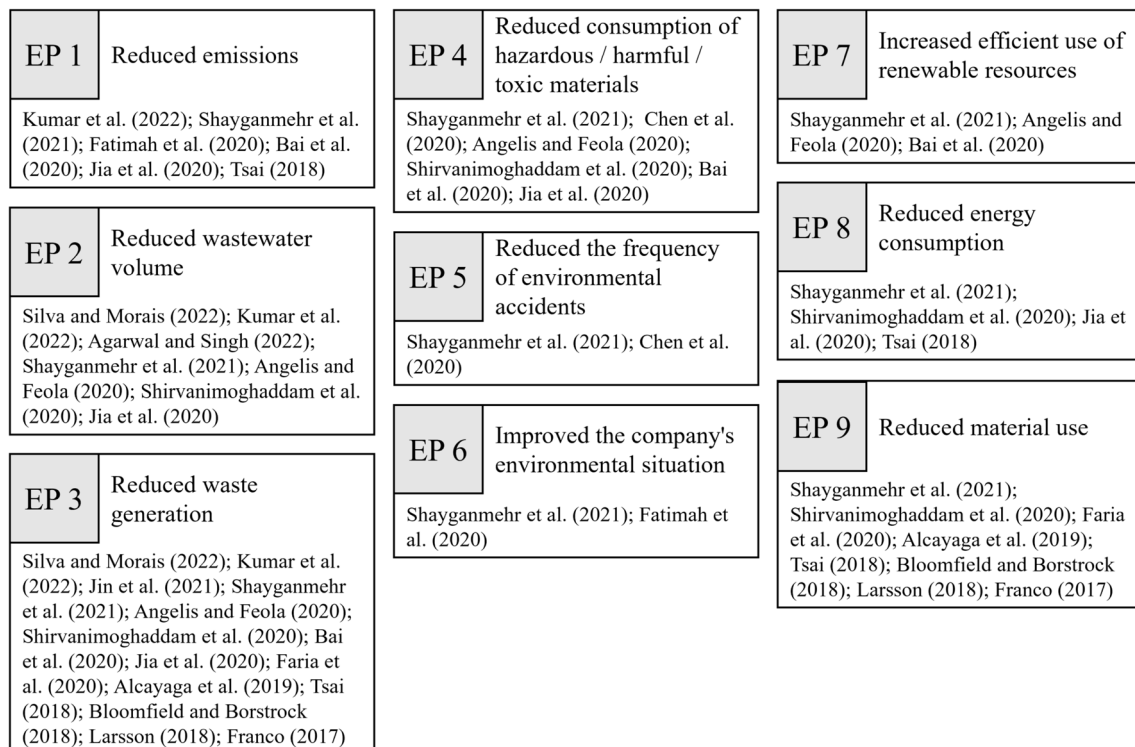


Fig. 7 Environmental performance



ECP 1	Reduced expenses with marketing materials Jia et al. (2020)	ECP 4	Reduced waste disposal fees Angelis and Feola (2020); Fatimah et al. (2020); Jia et al. (2020)	ECP 7	Reduced water and wastewater treatment costs Angelis and Feola (2020)
ECP 2	Reduced energy expenses Jia et al. (2020); Tsai (2018)	ECP 5	Reduced fines for environmental accidents Shayganmehr et al. (2021); Franco (2017)	ECP 8	Improved return on investment Shayganmehr et al. (2021)
ECP 3	Reduced waste treatment fees Angelis and Feola (2020); Fatimah et al. (2020); Jia et al. (2020)	ECP 6	Reduced raw material costs Alcayaga et al. (2019); Tsai (2018)	ECP 9	Increased profitability Shayganmehr et al. (2021); Tsai (2018)

Fig. 8 Economic performance

SP 1	Increased the satisfaction of stakeholders (recyclers, reverse logistics service providers, seamstresses, designers, and cooperatives in general) Shayganmehr et al. (2021)	SP 5	Increased job creation in circular economy practices Bai et al. (2020)
SP 2	Reduced environmental impacts and risks for the public Shayganmehr et al. (2021); Angelis and Feola (2020); Shirvanimoghaddam et al. (2020)	SP 6	Improved occupational safety, reducing the number of accidents Shayganmehr et al. (2021); Fatimah et al. (2020); Shirvanimoghaddam et al. (2020); Larsson (2018)
SP 3	Improved knowledge about community needs and rights Shayganmehr et al. (2021)	SP 7	Improved occupational health, reducing the number of employees on sick leave Shayganmehr et al. (2021); Chen et al. (2020); Angelis and Feola (2020); Fatimah et al. (2020); Shirvanimoghaddam et al. (2020)
SP 4	Improved community health and safety because the environmental impact reduction by local textile industry Shayganmehr et al. (2021); Angelis and Feola (2020); Fatimah et al. (2020)	SP 8	Improved the organizational climate, reducing absenteeism Shayganmehr et al. (2021); Fatimah et al. (2020)

Fig. 9 Social performance

making. Angelis and Feola (2020) concluded based on a case study in Italy that the use of big data favored the decision-making process concerning the volume of artichokes that become waste in manufacturing, allowing to schedule the production volume of natural dyes for sale. Thus, it generated a reduction in the pollution of water bodies and the use of renewable resources to paint fabrics, reducing environmental risks for society/community and the occupational health of employees. It also reduced water and sewage treatment costs. Fatimah et al. (2020) mention that big data was used for decision making in smart waste management of

textile waste to reduce, recycle, reuse, recover, repair, and remanufacture. By doing so, it generated environmental benefits because it reduced the emission level and improved the environmental awareness of the company, and reduced waste treatment/disposal costs. Alcayaga et al. (2019) add that big data assists in recycling decision making because it shares information to recyclers about waste value restoration, and it also empowers/mobilizes customers to sort each type of textile items before disposal, reducing the dumping of textile waste into regular trash. Bai et al. (2020) underscored the significance of leveraging big data to enhance data precision



in decision-making processes aimed at substituting hazardous materials, thereby boosting the utilization of renewable and recyclable resources in the cotton production chain, and fostering environmental preservation through efficient resource management. Chen et al. (2020) mentioned about the use of big data to analyze and optimize the consumption of energy and chemical materials in textile production, avoiding the potential environmental accidents and reducing the operational risk. Shayganmehr et al. (2021) applied the Fuzzy Method in Big data analytics for more accurate and intelligent decision making in a large Iranian textile industry, aiming at: efficiency in facilities and use of energy and renewable materials and resource recovery with recycling. This has reduced water, energy and hazardous material consumption and minimized waste generation, as well as improved return on investment with disseminated corporate ethics to society.

About autonomous robots, which work with humans, enabling learning and decision making, and solving complex tasks. Bai et al. (2020) mentioned that autonomous robots is generally little adopted in the textile industry. Shayganmehr et al. (2021) concluded that strong interaction between humans and robots improves production efficiency and sustainability in the textile chain by promoting substitution of production inputs to renewable and recyclable ones. This approach mitigates waste generation, diminishes the use of hazardous materials, conserves energy, water, and resources, while optimizing operational costs for a positive return on investment. Oliveira Neto et al. (2022) stated that the implementation of autonomous systems contributes to the CECP, promoting resilience and regeneration in the textile chain using renewable and recyclable resources (energy, water, materials) and textile waste recovery.

Simulation enables precise adjustments within complex textile industry systems through the utilization of knowledge, information, and accurate estimations. Tsai (2018) emphasized the critical role of simulation in strategically orchestrating green production within the textile industry, aiming to optimize design in the production system for toxin elimination, energy and material conservation, increased utilization of recycled components, and energy recovery. Thus, it increased recycling, reduced material consumption, emissions and energy, and reduced costs and increased profits. Bai et al. (2020) found that drones and simulation are sustainable in the textile industry because the drone collects data from agriculture for fiber crops, while simulation shows prediction. Thus, it uses agricultural production design to increase revenue, eliminates the use of toxic substances, and uses renewable materials. By doing so, it reduced the consumption of hazardous materials and increased the efficient use of renewable resources. Agarwal and Singh (2022) developed simulation using fuzzy Delphi allowing selection of wastewater treatment techniques in the textile industry

to promote circularity of water and elimination of hazardous products. Thus, it reduced the volume of wastewater, in addition to using biological treatment process with membrane filtration for removal of hazardous products, being economically feasible, in addition to elimination of pollution in the treatment system and community. Silva and Morais (2022) developed a simulation to order the outsourcing strategies for textile waste management. A shared economy was adopted with those involved (waste manager and clients) to reduce disposal through reverse logistics and recycling. This intervention minimized the consumption of virgin materials, water, and energy while curbing waste generation, thereby fostering local employment opportunities for waste pickers. Kumar et al. (2022) applied simulation in textile finishing process that requires water recovery from effluent using CP and I4.0 T. This generated savings in energy consumption (1,441,500 kWh), water (21%), waste generated (8–15%) and GHG emissions (14,064 Tons).

The internet of things enables the connection between network of physical objects, environments, vehicles, and machines through embedded electronic devices for system integration aimed at repairing textile items to resell with low price, reusing materials (recycled fabrics, fiber), remanufacturing, and recycling. Thus, it has generated cooperation for repair and recovery by designers of textile products to increase the useful life, contributing to the reduction of materials and raw material expenses (Alcayaga et al. 2019). Franco (2017) concluded that the adoption of IoT promoted the traceability of textile products by making real-time information available to logistics employees about collection points, recyclers, location, and accessibility of waste. Thus, it was possible to generate recycling, reuse, repair, and remanufacturing, minimizing waste generation and the consumption of virgin materials. Jia et al. (2020) conducted a theoretical review in which found that the traceability of textile products allows structuring the reverse chain, which includes recyclers, customers, third parties, and mainly focusing on product design for redesign. This contributes to eco-efficiency by reducing the consumption of virgin materials (cotton, wool, polyester), promoting recycling and reuse in addition to the reduction in the rate and discharge of textile products. Faria et al. (2020) also concluded through systematic literature review that no mobile technology using IoT was identified to reuse and redesign clothing. Thus, it could adopt virtual textile catalogs to simulate the products for customers, in addition to establishing service provision of design/stylist for collection creation, cutting and modeling. This approach would decrease waste generation and virgin material usage, thereby enhancing environmental performance.

Bai et al. (2020) adopted IoT in cotton cultivation to speed up decision making in terms of reducing waste generation and pesticide use. This promoted environmental gains by



minimizing waste generation and consumption of hazardous materials. Fatimah et al. (2020) proposed architecture for intelligent waste management using IoT from waste collection, segregation, classification, reuse, and recycling. Thus, information is available in real time, integrating environmental awareness culture, improving occupational health and safety, minimizing emissions. Chen et al. (2020) add that the principle of green chemistry needs to be applied in the dyeing of textile items because it uses hazardous products. Thus, the adoption of IoT is a solution for sharing and integrating information with partners, enabling the selection of less aggressive production resources. With this, it would be possible to avoid environmental accidents and operational risk by exchanging hazardous materials for renewable and recyclable resources. Shirvanimoghaddam et al. (2020) concluded that the use of IoT generated environmental proactivity in operational decision making with real-time information in terms of waste reduction, recycling, and reuse of textile waste, including the sharing of information for recyclers and customers. Thus, it allows to reduce/eliminate waste generation, recover textile waste in other companies and repair products and resell with low price through virtual catalog, generating information for the community. Jin et al. (2021) add that the biggest hotspot is the circular business model for fast fashion, which values sharing fashion with use of IoT through virtual textile catalogs for customers. This allows the cooperation of other maintenance/refurbishment companies on textile items in sewing workshops to increase the product life and share by renting the service for the time of use. Thus, it reduces the consumption of materials and waste generation and promotes community social responsibility. Shayganmehr et al. (2021) conclude that IoT improves competitive advantage due to information sharing with stakeholders: recyclers, customers with the use of virtual catalog, other textile products maintenance companies to increase product life, designers to reform textile products, suppliers for the development and selection of renewable and recyclable materials. This has enhanced environmental and economic performance by reducing waste generation, materials, and energy usage, minimizing wastewater volume, curbing hazardous material consumption, and optimizing renewable resource utilization.

Cloud computing adoption facilitates data storage and sharing, managing information on clothing reuse and redesign to promote material circularity, thereby reducing virgin material usage and waste generation (Faria et al. 2020), allowing to store data on cotton crop, facilitating information and communication of decision makers, mainly by sharing information with stakeholders on sustainable cotton production in terms of mitigation of pesticide use and efficient use of water and energy, reducing emissions (Bai et al. 2020). Therefore, cloud computing enables making data available and sharing information in the textile chain, including

recyclers to restore the value of waste; customers to mobilize them to sort textile items before disposal, and suppliers to develop and select renewable and recyclable materials. Thus, it contributes to the reduction in waste generation and mitigation of fines for environmental accidents (Shayganmehr et al. 2021).

The implementation of additive manufacturing in textile firms is related to the development of prototypes to promote customized solutions for people in general and people with special needs, aiming the development of customized apparel avoiding the generation of waste in the purchase of non-adapted textile products (Bloomfield and Borstrock 2018), diminish waste generation rates in production processes and conserve energy in manufacturing through the implementation of 3D technology that aids the development of product and redesign, thus enhancing efficiency and diminishing energy consumption (Shirvanimoghaddam et al. 2020; Bai et al. 2020; Bloomfield and Borstrock 2018). In this context, additive manufacturing aims at customization in the sustainable development of textile items to promote circularity. It employs DfE principles in development of product and packaging to facilitate recycling, repair, remanufacturing, maintenance, and innovation for durable products while phasing out toxic substances, reducing the consumption of hazardous materials and products, increasing the rate of return on investment (Shayganmehr et al. 2021). Thus, 3D technology can assist production planning and control in eliminating waste generation in production and develop products and logistics packaging for reuse and recycling, in addition to promoting shared economy with suppliers for the use of logistics packaging come and go (reuse) to achieve circularity and regeneration (Oliveira Neto et al. 2022).

Augmented reality enhances textile industry processes by enabling users to access and interact with virtual information directly within their environment, improving design, optimization, and verification. Thus, it was proposed for on the job training of employees and visualization / participation and customization of the customer in the design of textile products focusing on the concern with the end-of-life (waste), making it possible to develop products for reuse / repair / recycling. With this, it reduces waste and material generation and the number of accidents at work, improving job safety due to training simulation (Larsson 2018). Bai et al. (2020) found that augmented reality was not utilized within the textile industry studied. However, it is suggested to prioritize the technological aspects focused on product design and production process, and it is important to use augmented reality in partnership with stakeholders to eliminate the use of toxic substances, reduce energy, water, and material consumption, in addition to cost reduction and optimal return on investment (Kumar et al. 2022).

The implementation of Physical Cyber Systems is related to the use of sensors, processors and actuators monitored by



computers, including other elements in the physical layer such as radio frequency identification tags and barcodes for identification through optical readers, being important to provide the traceability of post-consumer textile waste. Thus, it generates repair, reuse, recycling and remanufacturing, contributing to the reduction of material use and waste generation (Alcayaga et al. 2019). Bai et al. (2020) reported that sensors and actuators in textile production enabled the availability of data for managers, reducing waste and promoting reuse. By doing so, it improved energy efficiency and reduced emissions. Tsai (2018) supports the utilization of sensors in green production planning and control to minimize waste, enhance recycling, decrease materials, emissions, and energy usage, thereby fostering environmental benefits alongside cost savings and heightened profitability. As an example, presented by Angelis and Feola (2020) related to the use of artichokes for the manufacture of ecological dyes for the dyeing of textile products using automation with remote sensing, aiming to increase the efficiency of the product and the process. By doing so, it reduces waste generation and the rate and treatment/disposal of production waste. Franco (2017) and Jia et al. (2020) corroborate that textile traceability needs to be activated in the production and distribution network to promote end-of-life waste circularity. It is proposed to employ universal product codes, radio frequency identifiers, and barcodes. It would mitigate environmental impacts by minimizing material usage and waste generation. Fatimah et al. (2020) mention that to promote smart waste management, it is important to deploy sensors, camera, micro control, and mechanical technology to perform end-of-life textile waste defibration for reuse. Thus, employees should be trained in environmental education to reduce emissions and waste treatment and discharge rates. Shirvanimoghaddam et al. (2020) add that for wastewater treatment, membrane nano reactor is used to extract contaminants and promote closed cycle. With this, water consumption and wastewater volume are optimized. Also, with the adoption of sensors in production reduced the process temperature and saved energy. Kumar et al. (2022) complements to incorporate sensors for wastewater monitoring. This optimized 21% in water consumption and reduced costs due to reuse. Shayganmehr et al. (2021) concluded that it is important to invest in IT infrastructure to improve efficiency in facilities aiming for reduction: in energy and materials consumption, in accidents at work to obtain environmental gains and return on investment.

I4.0 T in textile firms requires cybersecurity systems to protect data from possible threats and failures that can cause business disruption, for example, it is important to secure logistical information about the traceability of post-consumer textile waste to promote recycling and reuse. With this, it reduces waste generation, minimizes the use of materials (Franco 2017), another example, is to adopt

cybersecurity system in the biochemical processes related to dyeing to replace hazardous products with renewable and recyclable resources. Thus, it reduces the consumption of hazardous products in production and improves occupational health and safety (Shirvanimoghaddam et al. 2020). Also, in sharing information with recyclers, virtual catalogs with customers and other textile products maintenance companies it is important to invest in systems to promote data security to improve the environmental situation of the company with return on investment (Shayganmehr et al. 2021). However, cybersecurity is little used in the textile industry (Bai et al. 2020).

Also, the use of artificial intelligence is important for decision making in the textile chain, especially for optimizing chemical uses using automated controllers in different stages of dyeing and planting cotton, improving pulse rinsing technology, and promoting reuse and recycling of textile products (Shirvanimoghaddam et al. 2020). It is emphasized that farmers use artificial intelligence for energy efficiency in terms of quality, cost savings and emission reduction (Bai et al. 2020). Thus, textile industries are developing computational models for autonomous decision making (Shayganmehr et al. 2021).

Hypothesis development

Textile industries are utilizing I4.0 T as a catalyst for attaining sustainability goals and facilitating the implementation of CECP to enhance product and service quality within ethical and sustainable performance (Shayganmehr et al. 2021). Shirvanimoghaddam et al. (2020) reported on the implementation of additive manufacturing can reduce the rate in waste generation in production processes, promoting circular textile economy. Jia et al. (2020) demonstrated IoT and cyber-physical systems' role in enhancing CECP in the textile and apparel industry. Faria et al. (2020) highlighted cloud computing and IoT aid for online commerce, particularly in mobile applications aligning with CECP. Tsai (2018) concluded Industry 4.0 Technologies have facilitated green production planning and control in the textile firms. Thus, the findings suggest significant investments by large textile companies in I4.0 T integrated with CECP principles and actions (Oliveira Neto et al. 2022). Luthra and Mangla (2018), Rajput and Singh (2020), and Shayganmehr et al. (2021) also corroborate the coupling of CECP with Industry 4.0. Therefore, the first hypothesis emerges:

H1: The adoption of Industry 4.0 technologies by large textile industries stimulates the emergence of circular economy practices at the micro level considering cleaner production.

Studies were also found to provide qualitative evidence of I4.0 T and CECP to generate improved environmental, economic, and social performance. Fatimah et al. (2020)



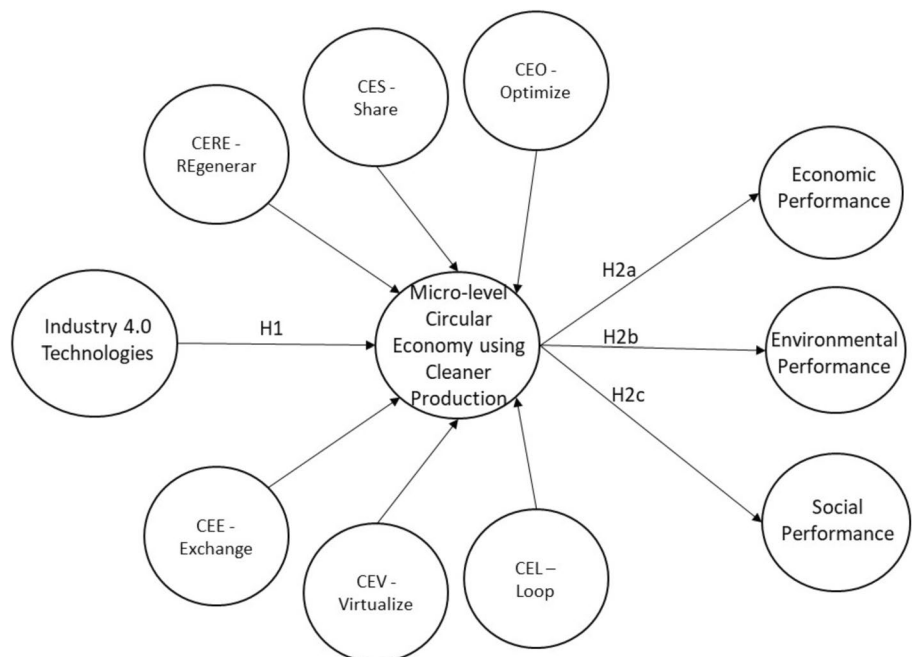
found that the use of I4.0 T with CECP in the textile sector an intelligent waste management system to enhance sustainability in Indonesia. Agarwal and Singh (2022) applied simulation to evaluate the performance of textile wastewater treatment techniques considering environmental aspects for reduction in wastewater volume, removal of hazardous products and water circularity, as well as, economic to apply economically feasible solutions and improvement in community health and safety due to elimination of pollution in the water system, contributing to social performance. Kumar et al. (2022) found that I4.0 T and CP reduced energy, water, and chemical waste across five textile production sites. Oliveira Neto et al. (2022) emphasized that the implementation of I4.0 T at LTLB enhanced CP through elimination in waste, emissions, and energy generation, promoting circularity in the production system. Jin et al. (2021) adopted IoT in the circular business model for fast fashion, reducing the consumption of virgin materials and waste generation, and promoting community social responsibility. Chen et al. (2020) mentioned based on green chemistry that the adoption of big data analytics with IoT optimized the consumption of energy and chemical materials in production, avoiding the potential environmental accidents, besides reducing the operational risk. Angelis and Feola (2020) showed a case in a circular startup in the Italian green paint market that adopted big data and cyber-physical systems to study the reuse of artichoke waste for paint application on fabric, and could reduce waste generation, rate and treatment/discharge of production waste, and water pollution. In addition to using renewable resources for fabric painting,

reducing environmental risks to society/community and occupational health of employees. Jia et al., (2020) proposed for future research to evaluate the sustainable performance of CE implementation in textile industry mainly valuing social performance. Thus, the second hypothesis is proposed:

H2a: The adoption of Industry 4.0 technologies with CE/CP generated improved economic performance; **H2b:** environmental and **H2c:** social.

The constructs of the conceptual model and the suggested hypotheses are shown in Fig. 10. The CECP construct was composed of the six strategies of the ReSOLVE framework: Regenerate (CERE), Share (CES), Optimize (CEO), Loop (CEL), Virtualize (CEV) and Exchange (CEE). Those strategies refer to constructs of the conceptual model, which were based on the definition of MacArthur (2015). The strategy Regenerate (CERE) proposes shifting to renewable energy and material, allowing energy and materials to circulate in a closed cycle; Share (CES) denotes for sharing economy, goods and assets are shared between individuals. With this, products should be designed to last longer, and maintenance should be available to allow reuse and extension of product life; Optimize (CEO) addresses to minimize residue in production process and across the suppliers; Loop (CEL) means to retrieve value of residue and restore the value of after consumption of products and packaging; Virtualize (CEV) is a service-focused strategy that changes the physical with virtual, dematerialized products; Exchange (CEE) approaches replacing non-renewable materials by renewable ones.

Fig. 10 Conceptual model and the suggested hypotheses



Materials and methods

The investigation described in this manuscript adopts a confirmatory research design, employing both qualitative and quantitative methodologies. Confirmatory research is chosen when a comprehensive theoretical understanding of a phenomenon has been established, utilizing precisely defined concepts, models, and propositions (Forza 2002). Following this section, the procedures used in the elaboration of the conceptual model and in the survey, research will be presented (Fig. 11).

The steps that make up the methods adopted in this work are presented in this illustration. It is worth noting that it is only possible to proceed to the next stage when the previous ones have been properly completed. Additionally, the execution of methodological protocols supports the increasing structural evolution of the work so that the results are presented correctly. Finally, the graphical presentation of methodological protocols also makes it easier for future studies to understand, repeat and improve these protocols.

Procedures for the elaboration of the conceptual model

The conceptual model proposed in this research consists of I4.0 T, CECP and sustainable performance constructs. The absence of a model with these characteristics indicated the need to survey the literature about the variables related to the mentioned constructs. Then, a systematic

literature review (Cassia et al. 2020) was conducted to identify the variables of the model, according to the beginning of Sect. "Literature review" of this paper.

The 18 selected articles formed the theoretical basis for building the conceptual model. The variables identified in these papers were: 10 from I4.0 T; 40 from CECP and 26 from sustainable performance. The 40 CECPs were broken down into six subgroups according to ReSOLVE framework strategies. The 26 sustainable performance metrics were also distributed into three subgroups: environmental, economic, and social performance.

Survey research

Survey research is a method used in quantitative studies to test hypotheses by means of statistical analysis. The results derived from survey research offer a quantitative portrayal of population trends, attitudes, or opinions (Creswell 2014). This methodology typically involves a structured procedure delineated in three key steps, as outlined by Forza (2002): (i) defining the target population to establish sample size requirements; (ii) executing a pilot study to assess the efficacy of the research instrument; and (iii) gathering field data from the predetermined sample and subjecting them to rigorous statistical analysis. Several papers have cited Forza (2002) as a relevant reference for survey research, such as Thomé et al. (2016), Buer et al. (2021), and Elia et al. (2021).

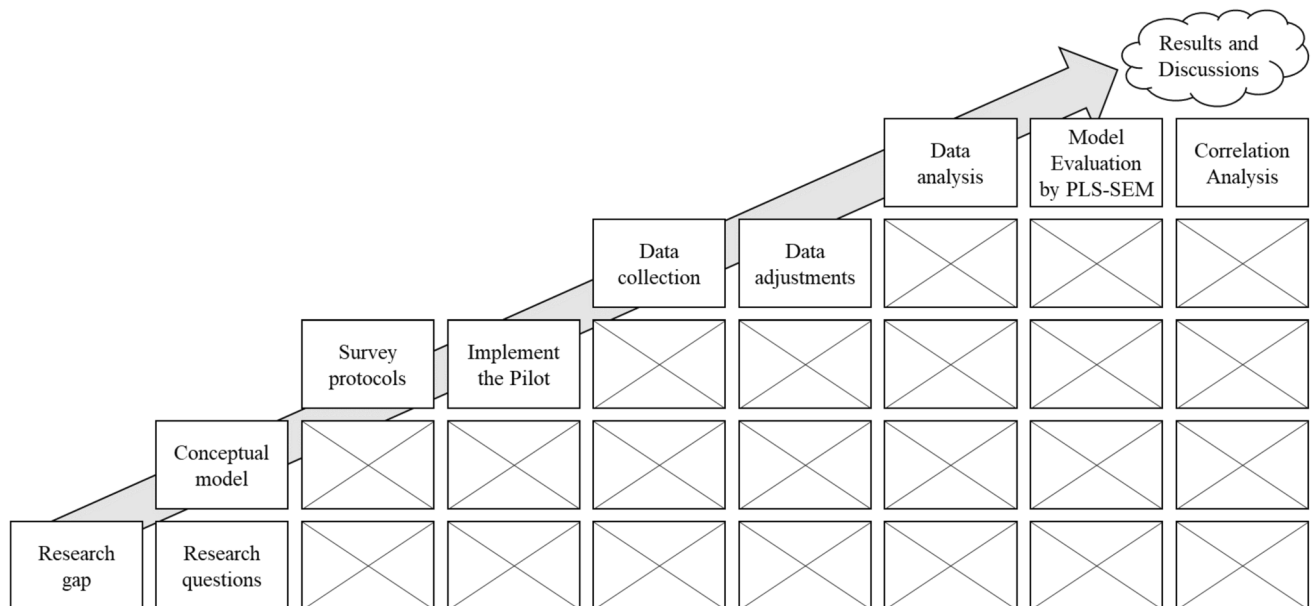


Fig. 11 Methodological protocols



Table 1 Discriminant validity of the model

	CEE	CEL	CEO	CERE	CES	CEV	ECP	EP	SP	TI4.0
CEE	0.9496									
CEL	0.6788	0.8356								
CEO	0.8320	0.6115	0.8710							
CERE	0.8590	0.6721	0.7671	0.8628						
CES	0.8489	0.5755	0.6599	0.7716	0.8833					
CEV	0.8021	0.5426	0.6259	0.7416	0.7766	0.8028				
ECP	0.7541	0.7446	0.6218	0.6866	0.6457	0.6654	0.7789			
EP	0.8045	0.5804	0.7508	0.6977	0.6683	0.5774	0.7340	0.8646		
SP	0.6308	0.4318	0.5004	0.5149	0.4928	0.5355	0.6473	0.6889	0.7585	
TI4.0	0.5891	0.6150	0.5721	0.5222	0.5937	0.6188	0.5950	0.4820	0.2632	0.7562

Table 2 Parameters of the adjusted model

	AVE	Composite Reliability	R ²	Cronbachs Alpha	Square root	Q ²	f ²
CE	0.5558	0.9799	0.4564	0.9783	0.7455	0.2250	0.4891
CEE	0.9017	0.9649	0.8986	0.9453	0.9496	0.7400	0.7435
CEL	0.6983	0.9585	0.6433	0.9418	0.8356	0.4058	0.5729
CEO	0.7586	0.9658	0.7593	0.9400	0.8710	0.5222	0.6484
CERE	0.7445	0.9356	0.8196	0.9139	0.8628	0.5627	0.5270
CES	0.7802	0.9696	0.7817	0.9447	0.8833	0.5569	0.6577
CEV	0.6445	0.8750	0.6691	0.8159	0.8028	0.3703	0.4441
ECP	0.6066	0.9325	0.6158	0.9183	0.7789	0.3322	0.4684
EP	0.7475	0.9638	0.6135	0.9376	0.8646	0.4084	0.6486
SP	0.5753	0.9146	0.3372	0.8946	0.7585	0.1575	0.4589
TI4.0	0.5719	0.9281		0.9131	0.7562	0.4877	0.4877

Pilot study implementation

A form with a set of assertions about the observed variables was submitted to 10 specialists in the textile industry as a pilot study to validate the research instrument. Pilot study is a necessary step in survey research to adjust and validate the research instrument before its application in the field (Forza 2002). The role of pilot testing is ensuring the effectiveness of survey instruments (Dillman et al. 2014). A preliminary field pretest of the instrument and procedures is essential to assess the functionality and efficacy of data collection protocols and survey instruments under real-world conditions (Fowler 2014). The questionnaire variables adopted for this research are shown in Table 1 and 2.

Data collection and analysis

Sampling and data collection are crucial for obtaining a representative sample of the target population (Groves et al. 2009). The questionnaire, centered on the study's focal theme, was distributed to 150 managerial professionals specializing in operations and environmental management within prominent textile industries. This distribution

occurred in the second half of 2022. A total of 102 completed forms were collected in response to the survey. The minimum sample size was determined using G*Power 3.1.9.2 software, with parameters set to a mean effect size (f^2) of 0.15 and a test power of 0.80, according to the procedure recommended by Cohen (1988) and Westland (2010). The result of the calculation indicated that at least 55 responses should be obtained.

The gathered data underwent analysis of the conceptual model to elucidate the interrelationships among the constructs and their respective variables. This investigation employed Structural Equation Modeling (SEM) to analyze the data. SEM, which gained prominence through extensive research in the 1970s, is recognized for its confirmatory nature. However, it is noteworthy that SEM imposes certain constraints, such as requirements for substantial sample sizes and assumptions of data normality (Kline 2015; Schumacker and Lomax 2015; Hair et al. 2017a, b).

Moreover, the employment of the Partial Least Squares (PLS) technique has facilitated the estimation of stable regression coefficients, thereby mitigating the influence of insignificant variables on model calibration (Henseler et al. 2015). PLS presents inherent limitations in its application



and in extrapolating results, potentially leading to an overestimation of its capabilities (Vandenberg 2006; Rönkkö et al. 2016). Hence, it is imperative to judiciously justify the adoption of PLS (Rönkkö and Evermann 2013). To address these limitations, researchers have increasingly turned to the integration of PLS-SEM, as evidenced by a growing body of literature (Rönkkö et al. 2016; Hair et al. 2017a, b). However, the application of PLS-SEM mandates careful scrutiny and substantiation (Hair et al. 2019). It is advocated that model constructs rely on composite indicators rather than solely on theoretical concepts (Sarstedt et al. 2016). Ensuring the quality of PLS-SEM outcomes necessitates a profound understanding and adept application of the methodologies and metrics detailed in existing literature (Hair et al. 2016; Sarstedt et al. 2016).

Aware of the limitations, this study employed SmartPLS software to construct a PLS-SEM structural model, facilitating rigorous statistical analysis and elucidation of variable interrelationships. Circles within the model denote constructs, while rectangles signify variables. The connections between constructs delineate research hypotheses, quantified as path coefficients (Henseler et al. 2015). To uphold methodological rigor, a systematic step-by-step approach was meticulously adhered to, as outlined in the following.

The initial step involved conducting a statistical test to evaluate the normality of the collected data distribution. A distribution was considered normal if Skewness values fell within the range of -1.0 to $+1.0$, and Kurtosis values were less than 2.0 . Subsequently, model fitting was carried out following a methodology outlined by Silva et al. (2021), comprising eight steps:

- (i) Convergent validity of the model was assessed by ensuring that all Average Variance Extracted (AVE) values exceeded 0.50 .
- (ii) Discriminant validity was established if Pearson correlation coefficients were lower than the square roots of the AVEs.
- (iii) Model reliability was confirmed when Composite Reliability (CR) and Cronbach's Alpha (CA) values exceeded 0.70 .
- (iv) Significance of correlations and regressions was determined by ensuring that the t -test coefficient of Student equaled or exceeded 1.96 .
- (v) The Pearson coefficient (R^2) was utilized to gauge the effect of one construct on another, with 0.26 representing a large effect, 0.13 a medium effect, and 0.02 a small effect.
- (vi) Predictive validity (Q^2) was assessed by ensuring its value exceeded zero, and the effect size (f^2) was deemed large for values near 0.35 , medium for 0.15 , and small for 0.02 .
- (vii) Model fit accuracy was evaluated by confirming a positive value for the Stone-Geisser indicator.
- (viii) With the completion of the preceding steps, interpretation of the calculated path coefficients became feasible.

Following the hypotheses testing, Pearson's correlation test was employed to ascertain the association capacity and linearity among the variables. This validation enabled the consideration of factor loading values in the correlation between the constructs of the model and their respective variables. The Pearson correlation test facilitates the examination of the relationship between two elements, yielding results ranging between -1 and $+1$, denoting negative or positive correlations of varying intensity (Chen and Popovich 2002; Pallant, 2016). The assessment of relationship intensity adhered to the criteria set forth by Dancey and Reidy (2007): weak correlation falls within the range of 0.10 to 0.30 , moderate correlation spans from 0.40 to 0.60 , and strong correlation is observed above 0.70 .

Assessment of the measurement model

This section presents the unfolding of the steps described for model fitting, the analysis of the path coefficients, and then the decision to accept or reject the research hypotheses. The analysis of the Skewness and Kurtosis values indicated that the data are in a normal distribution.

Model fitting began with the removal of variables with factor loadings and AVE values less than 0.50 (Henseler et al. 2015). A new round of evaluation was performed for each extracted variable because the factorial loadings changed due to the change in the model. Then, the variables were extracted one by one until only factor loadings and AVE values greater than 0.50 remained. Only the cloud computing variable (T_5), from the industry 4.0 technology construct, had a value less than 0.5 . The large textile industries considered the investment in cloud computing irrelevant due to the difficulties with information security. According to the specialists, it would be necessary to improve cyber control, and without this control, inattentive employees could open gaps for hacker invasions or data leakage. Removing this variable ensured the convergent validity of the model. The adjusted measurement model with the factor loading values is shown in Fig. 12.

The subsequent phase involved evaluating the discriminant validity of the model through cross-load analysis (Kline 2015). This analysis entails ensuring that the square root of the AVE exceeds the correlation value associated with the respective construct (Hair et al. 2019). The AVE square root values, shown on the main diagonal of Table 1, are higher than the other values in the column, which refer



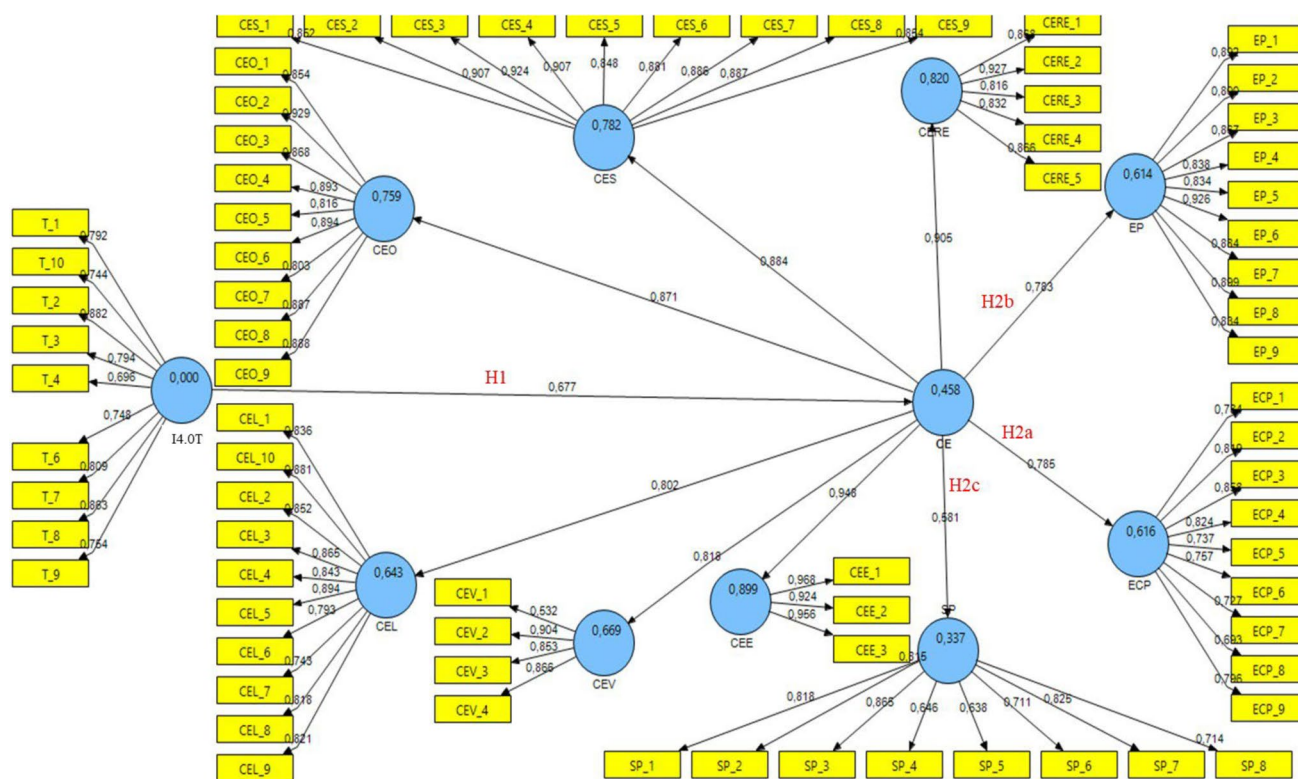


Fig. 12 The adjusted measurement model with the factor loading values created in SmartPLS

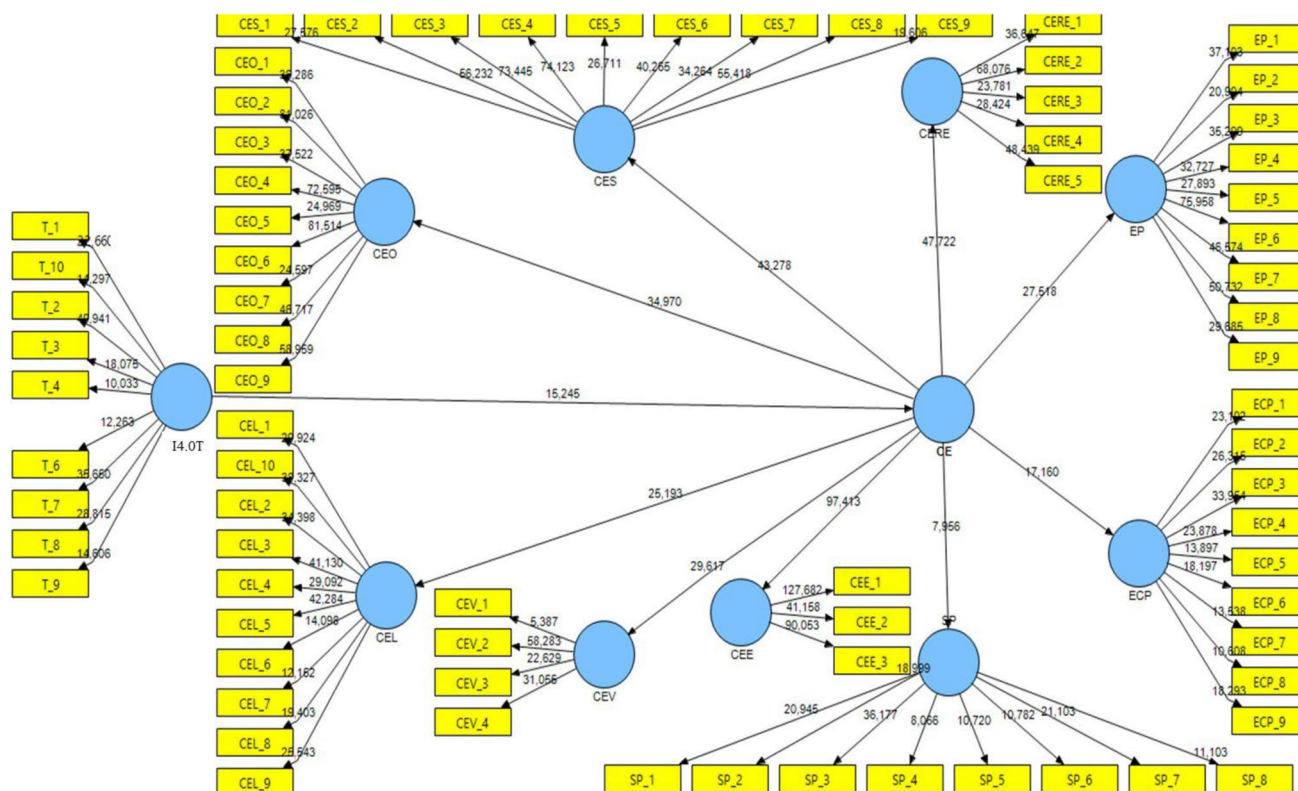


Fig. 13 Result of the t-test of student run in SmartPLS bootstrapping



to the correlation of the constructs. This result denotes the discriminant validity of the model.

The subsequent step involved evaluating the reliability of the model, employing Composite Reliability (CR) and Cronbach's Alpha (CA) as criteria, with a threshold set at 0.70 (Hair et al. 2012). The values shown in Table 2 indicate that CR and CA meet this criterion in all variables.

The analyses advanced to evaluate the statistical significance of correlations and regressions (Hair et al. 2012). The t-test of Student was run on the SmartPLS bootstrapping function. Value of t greater than 1.96 indicates that correlation and regression are significant. The bootstrapping result indicated that all t-values are greater than 1.96, as shown in Fig. 13.

Coefficient of determination (R^2) of Pearson, calculated next, indicated the effect of one construct on the other. Cohen (1988) suggests that R^2 greater than 0.26 denotes a large effect. The R^2 values are shown in Table 2, all greater than 0.26. The construct TI4.0 has no precedent, that is, it receives no effect from another construct. For this reason, the R^2 of the IT4.0 construct is blank in the table.

The accuracy of the model fit was checked using the Stone-Geisser indicator (Q2). All Q2 values were positive. Indicator (f2) of Cohen represents the effect size of the construct on the model fit. All f2 values were greater than 0.35, which denotes a large effect of the constructs in the model. Thus, according to Hair et al. (2012) criteria, the predictive validity of the model was confirmed. The results of Q2 and f2 are shown in Table 2.

The last step of model evaluation was the test of Pearson, which was intended to verify the correlation between variables of different constructs. SmartPLS is limited in this regard and does not offer this function, which restricts the interpretation of the results (Vandenberg 2006; Rönkkö et al. 2016). Following the criteria proposed by Dancey and Reidy (2007), only correlations with a value equal to or greater than 0.50 were considered significant (Fig. 14). The results of test of the Pearson are presented in full in the appendix of this paper.

Results and discussion

This section presents the discussion of the results of the adjusted measurement model that was shown in Fig. 2. The path coefficient between the TI4.0 and CECP constructs, whose value is 0.677, confirms that H1 should be moderately accepted. Thus, IT4.0 T deployed by LTILB moderately drives the adoption of CECP. With this, it shows that LTILBs have started investing in I4.0 T, which contributes to CECP. Research that relates I4.0 T, CECP and performance usually only mentions scientific evidence through exploratory study, usually using case studies and systematic literature review. Thus, this study is the first to measure the relationship between IT4.0 and CECP, in a confirmatory way, in LTILB, bringing relevant results about this junction. Also, this study is important to encourage managers to adopt I4.0 T in face of the opportunity to contribute with the environment, being an important competitive aspect to compete in the European market. The adoption of I4.0 T with CECP in a country promotes knowledge for society, making it possible to generate new jobs for qualified labor.

A relevant aspect was that 4 subgroups of ReSOLVE that compose the CECPs measured above 0.8, but the main highlight found were CEE and CERE with path coefficient above 0.9. MacArthur et al. (2015) mention that CEE represents the Exchange aimed at replacing old, non-renewable assets with renewable ones in the production system and CERE, which considers regeneration in terms of switching to renewable energy and circularity of waste and materials in closed cycle in production. Oliveira Neto et al. (2022) stated that large textile firms are more resilient in adapting CE practices in the production system than small and medium-sized enterprises. Silva and Morais (2022) emphasized the challenges faced by small and medium-sized enterprises (SMEs) in achieving alignment with the CE principles. Kumar et al. (2022) highlighted the proactive measures undertaken by smaller industrial facilities to mitigate water consumption during manufacturing operations, emphasizing a commitment to reuse wherever feasible. This finding contributes to the literature and organizational practice because it shows that LTILBs are changing to replace non-renewable production resources with renewable ones, which include energy use, material from renewable sources, as well as providing circularity of textile product waste or from industries with similar waste. Moreover, the implementation of I4.0 T besides stimulating CECP generated strong positive impact on performance H2a: economic (0.785); H2b: environmental (0.783) and moderate on H2c: social (0.581). Thus, it is concluded that LTILB that adopted I4.0 T have stimulated CECP focusing only on economic and environmental,

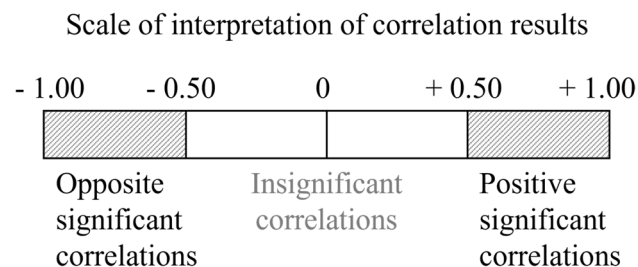


Fig. 14 Criteria for interpreting pearson test results



according to de Paoli et al. (2013) and Oliveira Neto et al. (2015) gain because they aim to improve competitiveness in the foreign market, forgetting to obtain relevant social results, such as job generation to reduce hunger, better communication for the community, and reduction of environmental impacts for the community.

After hypothesis testing, correlations between variables of I4.0 T and CECF were established using Pearson's correlation coefficient. The implementation of cyber-physical system in production was the most relevant (T8:0.883), according to the charge in Fig. 2, which stimulated (CEV_4-EP_3) the implementation of digital services for system setup, coupled with online services for devices maintenance, yielded a proactive reduction in waste generation within the production system. This approach effectively mitigated scrap occurrences, contributing to enhanced operational efficiency and resource utilization. Larsson (2018) stated that advancements in sustainability and cutting-edge technology are progressively influencing and steering innovation within the apparel and fashion sector, which includes the integration of 3D visualization tools for virtual representation and fitting, as well as the adoption of digitally enhanced manufacturing tools in sales and production processes. Jin et al. (2021) asserted that textile industry has carried out business model sustainable innovation influenced by emerging technologies such as virtual reality. Shayganmehr et al. (2021) conducted a case study within the Iranian textile industry, highlighting the imperative need for the integration of IT infrastructure, the implementation of digital technician services and online support mechanisms for the maintenance of machinery and equipment. However, this study admitted that the adoption of the cyber-physical system in the textile industry contributed to the evolution of the digital service with online technical support for real-time monitoring of the processes, improving the interaction between the stages of production, resulting in traceability, with reduction in the volume of scrap generated in the process of spinning and weaving. It is noteworthy that real-time monitoring allows the identification of anomalies before generating rejects in the dyeing and manufacturing processes, in addition to reducing the cycle time of the chain, which minimizes the consumption of production resources and increases the accuracy of information about the stages of production processes, from the production of fibers/filaments, through the production of yarn, fabric, until the transformation into finished pieces.

An additional significant discovery pertained to the integration of autonomous robots into textile manufacturing (T_2:0.882), yielding a notable reduction in production inputs such as raw materials, water, and energy. This adoption consequently facilitated an augmentation in the utilization of renewable and recyclable resources (CEE_1-EP_3-EP_8-EP_9-EP_7-EP_6-SP_2-SP_3), having a positive effect on environmental performance because it

reduced waste generation, energy consumption, material use, increasing the efficient use of renewable resources, which consequently improved the environmental situation of the company, generating improvements in social performance because it reduced environmental risks for the general public by disseminating knowledge about the needs and rights of the community. Oliveira Neto et al. (2022) mentioned about autonomous robots in an exploratory way, indicating about the possibility of reduction in waste generation, energy consumption, and material use. In this context, this study confirmed that LTILB uses autonomous robots in production, from tasks such as: feeding machines, transporting materials, identifying, and segregating textile materials, performing more complex activities (splicing, sewing, applications, making whole pieces) and currently in the process of recycling textile products, reducing waste generation and the consumption of raw materials. Also, the adoption of autonomous robots in the production of whole pieces reduces energy and water consumption, besides reducing ergonomic problems and the exposure of employees to unhealthy and dangerous environments. Shayganmehr et al. (2021) emphasized that employing robots is aimed at streamlining human-machine interaction, expediting the execution of CECF practices, with the overarching goal of reducing resource consumption through enhanced efficiency. Shirvanimoghaddam et al. (2020) found that the implementation of automated dosing machines leads to a significant decrease in the utilization of chemical inputs, highlighting their efficacy in optimizing resource allocation and fostering sustainability across diverse industries.

The incorporation of autonomous robots into textile production (T_2:0.882) integrated suppliers to develop and select renewable and recyclable materials (CEE_3-EP_3-EP_9-EP_7-SP_3-SP_2). This collaborative effort has influenced positive effects on environmental performance, primarily manifested in the reduction of waste generation and a more judicious use of materials, thereby optimizing the utilization of renewable resources. By engaging suppliers in the search for renewable and recyclable materials, the adoption of autonomous robots mitigates environmental risks for society at large, which contributes to leverage social performance. It is worth mentioning that a scarcity of existing research was identified regarding the linkage between the adoption of autonomous robots and the integration of suppliers in the selection of sustainable materials. This fact neglects the benefits reached through waste reduction and management, which aids in the mitigation of both environmental and social impacts. Further investigation is warranted to fully elucidate the myriad of positive outcomes stemming from the integration of autonomous robots in textile production, with a particular emphasis on waste elimination. This necessitates future research efforts to comprehensively explore and understand the broader spectrum



of benefits associated with this integration, including evaluating the extrapolation of results from industries with similar characteristics.

As a result, the integration of autonomous robots in textile manufacturing (T₂:0.882) has facilitated the substitution of hazardous materials with renewable and recyclable resources. This transition includes utilizing by-products from alternative manufacturing processes as raw materials for new product fabrication. Furthermore, natural pigments derived from waste materials, such as artichoke residue, are employed as alternatives to chemical paints, further enhancing sustainability in the production process (CEE₂-CERE₅-SP₃) improving social performance, generating knowledge about the needs and rights of the community. It is noteworthy that the substitution of hazardous chemical materials reduced/eliminated the emission level in the production system and local community (CEO₈-EP₁), generating relevant environmental performance. Although there is a lack of existing research on this topic, the absence of studies can be attributed to the comprehensive integration of autonomous robots into the production process. This integration is marked by a choice of utilizing renewable and recyclable materials, thereby eliminating the use of hazardous substances, particularly during the dyeing process. The careful selection of materials has not only eradicated the need for workers to handle these health-threatening elements but has also mitigated emissions within the production system and the surrounding local community. This eco-friendly approach extends beyond the production facility, actively disseminating its positive impact throughout the local community.

The adoption of an autonomous robot in textile production (T₂:0.882) also resulted in regeneration actions, focusing on the efficiency of the facilities because it applied technological improvements and started using renewable energy (CERE₂-EP₈-EP₇-EP₉-EP₅-EP₆-SP₂-SP₃) generating positive impact on environmental performance because it reduced energy and material consumption, increasing the efficient use of renewable resources. By doing so, it minimized environmental accidents and improved the environmental status of the company, reflecting positively on social performance because it made public about community rights on environmental risk mitigation. Shayganmehr et al. (2021) concluded through case study that human–robot interaction improved eco-efficiency in Iranian textile chain by reducing energy consumption. However, this confirmatory study concluded that the deployment of autonomous robots in the recycling processes in LTILB production has become more efficient and less dependent on non-renewable energy, shifting to consuming regenerated textile waste, e.g., the defibbing of made-up items, reducing the carbon footprint for society, as well as minimizing ergonomic and health risks.

Augmented reality (T₂:0.807) is also contributing to the Exchange for CEECP, which simulates production before applying it to dyeing and apparel factories, generating projections for production managers, stylists, and process engineers to generate less waste. Thus, this process facilitates the substitution of production inputs such as raw materials, water, and energy, as well as hazardous materials, thereby enhancing the utilization of renewable and recyclable resources (CEE₁-CEE₂-EP₃-SP₂). With this, it favors environmental performance by reducing waste generation and contributes to social performance by reducing environmental impacts and risks to society. Bai et al. (2020) did not identify the adoption of augmented reality in textile production but suggest the need to prioritize the use of this technology to promote CE. Kumar et al. (2022) developed case study and simulation in Pakistan and suggest the use of augmented reality in production and in partnership with stakeholders to eliminate the use of toxic substances, reduce energy, water and materials consumption, allowing cost reduction. Thus, the augmented reality connects the tangible and digital fields, providing multiple applications within the textile production sector. Its utility extends beyond the involvement of technicians with expertise in maintenance and operations. Augmented reality plays a pivotal role in the theoretical aspects, and it yields practical outcomes for organizational operations. Augmented reality facilitates the collaboration of technicians, enabling them to enhance production efficiency and minimizing waste. An example of its practical application lies in the ability to share virtual models beforehand. This enables a preview of designs prior to their implementation in processes such as dyeing and apparel manufacturing.

In addition, the adoption of augmented reality (T₂:0.807) stimulated the use of design for environment in product development and packaging allowing recycling, repair, remanufacturing, maintenance. The study additionally unveiled novel attributes enhancing the durability of products while simultaneously eradicating the utilization of hazardous substances (CEO₂-EP₃), minimizing waste generation, benefiting environmental gain because it optimized waste in production. Larsson (2018) applied action research in Sweden and concluded that augmented reality enables customer participation and customization in product development, incorporating packaging strategies designed to enhance the circularity of textile waste at the end of its lifecycle, reducing waste and material generation. Bai et al. (2020) stated that it is important to adopt augmented reality in product development to contribute to EC. The integration of augmented reality (AR) within the textile industry represents a significant advancement, fostering innovation in product design through the facilitation of mold development, which are virtually forwarded to production, thus reducing the development of physical prototype (dematerialization),



mainly generating packaging reduction. Then, this technological integration not only adds theoretical value but also enhances operational practices within the textile industry or even in industries with similar characteristics.

Another relevant I4.0 T used was big data analytics (T₃:0.792) using multivariate statistical evaluation and machine learning techniques to Exchange for EC. Thus, studies were conducted to replace productive inputs (raw material, water and/or energy) with renewable and recyclable resources (CEE_1-EP_3-EP_8-EP_9-EP_7-EP_6-SP_2-SP_1-SP_3-SP_7), resulting in environmental and social gains by reducing waste generation, energy consumption, and material use, which consequently improves the use of renewable resources and the environmental situation of the company. It also found a reduction in environmental risks to society and stakeholder satisfaction, as well as an improvement in occupational health, reducing the number of employee leaves of absence. It is emphasized that with the use of Big data analytics (T₃:0.792) to handle production data, it was feasible to design Exchange for the seamless integration of suppliers in the sourcing and selection process of renewable and recyclable materials (CEE_3-EP_7 > EP_9), which increases the adoption of renewable resources and reduces the use of materials, contributing to the environmental benefit. Bai et al. (2020) emphasized in their exploratory studies that it is important for big data to handle data more accurately for decision making in terms of replacing hazardous inputs and materials to enhance the utilization of renewable and recyclable resources within the textile industry. Chen et al. (2020) emphasized that integrating IoT and Big Data offers a novel solution for eco-industrial parks and suppliers, aimed at reducing environmental impacts and costs while optimizing energy, resources, and service quality. The findings of this research corroborate the insights provided by the mentioned studies. This research complements existing theories, validating the fundamental role of Big Data analysis in supporting decision-making processes in textile production, performing in the same way as industries that already use big data analysis on a consolidated basis. By highlighting the power of Big Data analytics, this study affirms its capacity to provide more reliable information, thereby facilitating the transition from non-renewable to renewable productive resources. This emphasis on data-driven decision-making serves as a catalyst for fostering a more environmentally conscious and resource-efficient approach.

Also using big data analytics (T₃:0.792) recognizing the lack of capacity for textile waste recyclability a business unit was designed to support reverse logistics and promote recycling and increase company control over the product life cycle (CES_9-EP_9), which would reduce the use of virgin materials favoring the environment. Research by Fatimah et al. (2020) and Alcaayaga et al. (2019) highlighted the role

of big data analytics for textile product recycling decisions. On the other hand, no study mentioned about reverse logistics. Thus, the use of big data analytics to design reverse logistics allowed closing the cycle, connecting customers to properly dispose of waste, as well as encouraging recycling and reuse, maintaining circularity and reducing resource consumption. In addition, fashion is the main generator of irregular textile waste/disposal, requiring reverse logistics for intelligent recycling with a focus on circularity at the micro level.

It is emphasized that for effectiveness in big data analytics, it is important to establish simulation (T₃:0.794) in the production area to reduce/eliminate emissions (CEO_8-EP_3-EP_8-EP_7), finding positive environmental outcomes through waste reduction, energy efficiency, and enhanced utilization of renewable resources. Bai et al. (2020) developed case studies and found that drones have the potential to collect data from agriculture for fiber crops, while simulation enables the analysis of this data for prediction. Tsai (2018) simulated design for the environment in textile production to increase revenue. Silva and Morais (2022) simulated circular strategies for outsourcing textile waste management in Brazil. Agarwal and Singh (2022) and Kumar et al. (2022) simulated wastewater treatment techniques in the textile industry to promote circularity/water use and hazardous product disposal. Despite finding these articles that use mathematical modeling, no research was identified that developed simulation using machine learning techniques. Thus, it was confirmed that the application of simulation enables new discoveries, interpretation, and communication of data, facilitating insights for effective decision making. With this, prediction of actual production is developed for the optimization of production resources (materials and energy) of the spinning, weaving, knitting, dyeing, and apparel processes.

Furthermore, despite limited utilization, artificial intelligence (T₁₀: 0.744) aided in forecasting the substitution of productive inputs to enhance the proportion of sustainable resources (CEE_1-EP_3-EP_8), predicting the opportunity for reduction in waste generation and energy consumption, contributing to environmental performance. Two case studies mentioned about simulation and artificial intelligence in cotton farming (Bai et al. 2020) and in resource optimization in textile production (Shayganmehr et al. 2021) and a literature review suggested the use of artificial intelligence for replacing chemical dyes with natural dyes (Shirvanimoghaddam et al. 2020). In this context, this study confirmed that artificial intelligence will be more effectively adopted for the development of autonomous systems. Specifically, artificial intelligence applications are helpful in optimizing fiber planning and consumption, leading to reductions in waste and mitigating the adverse impact on the utilization of renewable resources. This underscores the benefits



of integrating artificial intelligence technologies across domains, showcasing its potential to drive sustainability and resource optimization, contributing to Pinto et al. (2020).

Additive Manufacturing (T₆:0.748) and Cyber Security Systems (T₉:0.754) despite having a load greater than 0.7, being important for stimulating CECP, showed no correlations. This result corroborates the finding of Bai et al. (2020) that showed low overall performance of additive manufacturing and cybersecurity. In terms of additive manufacturing, there is a lack of operational strategy to implement this technology in textile production, needing to be better explored by the apparel chain, with a focus on customization and personal style (Bloomfield and Borstrock 2018; Oliveira Neto et al. 2022). The additive manufacturing needs to be targeted for textile waste reduction in fashion industries (Shirvanimoghaddam et al. 2020). Additive manufacturing significantly enhances operational efficiency and expedites the implementation of CECP (Shayganmehr et al. 2021). Another relevant aspect, although the use of Cyber Security systems (0.754) is important for data protection, especially in the transition to digital transformation, textile industries do not recognize the possibility of promoting CECP actions and sustainable performance. For example, to provide information security with encrypted data for textile waste traceability (Franco 2017), adoption of secure biochemical processes using cybersecurity system for hazardous product substitution (Shirvanimoghaddam et al. 2020) and promote security in customer data from textile virtual catalogs (Shayganmehr et al. 2021).

Finally, the IoT (T₄: 0.696) stimulated actions of the Exchange and virtualization of CECP. Thus, it replaced productive resources (raw material, water and or energy) for renewable and recyclable resources (CEE₁-EP₆) improving the environmental situation of the company, as well as, allowed to integrate suppliers for development and selection of renewable and recyclable materials (CEE₃-EP₇), increasing the efficient use of renewable resources. IoT has a focus on providing: real-time communication with operational managers about production resource consumption (Bai et al. 2020; Shirvanimoghaddam et al. 2020), cooperation and information sharing with designers and couturiers (Alcayaga et al. 2019; Chen et al. 2020), end-of-life textile waste traceability (Franco 2017; Jia et al. 2020), clothing reuse and redesign (Faria et al. 2020), smart waste management (Fatimah et al. 2020), and supplier monitoring for sourcing and selection of renewable and recyclable materials (Shayganmehr et al. 2021). Nevertheless, existing studies employing multivariate analysis have not yet to conclusively validate that the integration of Internet of Things (IoT) technologies within the textile industry has led to the replacement of conventional production inputs

with renewable alternatives. Moreover, empirical evidence is insufficient to substantiate the assertion that implementing IoT in this sector streamlines the procurement of renewable materials from suppliers. Additional research is required to thoroughly explore these potential connections and establish a more comprehensive understanding of the intricate dynamics between IoT adoption, production inputs, and the sourcing of renewable materials within the textile industry, even as few companies achieve this level of synergy with this common objective.

Also, they implemented IoT (T₄:0.696) for system configuration in digital service for machine maintenance (CEV₄-EP₃-EP₄-ECP₃-ECP₄), reducing waste generation, consumption of hazardous materials, and fees for waste treatment and disposal. Some research adopted IoT for customer interaction through virtual catalog (Jin et al. 2021; Shayganmehr et al. 2021). Another finding was that IoT adoption (T₄:0.696) related to economic performance, recognizing that CECP virtualization actions reduce fees for waste treatment and discharge. It is emphasized that although economic performance shows strong influence on CE actions in global data analysis, it only correlated with IoT. No studies were found referring to the application of Internet of Things (IoT) in digital technician services, particularly for online maintenance support in spinning and weaving mills. This technology could be useful to access real-time data, facilitating swift and informed decision-making in the upkeep of machinery and equipment within these industrial settings.

Conclusion

It is concluded that this is the first study to confirmatively assess the relationship between I4.0 T, CECP and sustainable performance in LTILB, denoting relevant contribution to theory and practice. The implementation of I4.0 T by LTILB encouraged moderately the CECP actions; however showing a change profile, which can be justified considering that I4.0 T stimulated the actions in production related to exchange of productive inputs and hazardous materials for renewable and recyclable resources, energy and materials regeneration, optimization/elimination of emissions and toxic substances through product design for environment, and virtualization of online service for machinery and equipment maintenance. On the other hand, there were no CECP actions aimed at sharing and cycle, denoting the absence of collaborative economy among companies for creating durable, reusable products, and the lack of a circular process for reclaiming waste and restoring post-consumer product value.



Thus, the implementation of I4.0 T by LTILB moderately stimulated the CECF actions not validating the sustainable performance, because it was concluded focus on economic and environmental gain aimed at improving competitiveness in the foreign market. It is noteworthy that in the competitive scenario, LTILB are pressured by the European market to improve the environmental situation of the company, and thereby reduce the use of energy and materials and waste generation, increasing the use of renewable resources. They are also pressured by Chinese industries to reduce the price of their products. With this, they work on reducing operational costs, reducing fees for discharge and waste treatment, and consequently improving eco-efficiency in product development and production process. This reduces environmental risks for the community and improves social aspects such as generating knowledge and increasing the level of satisfaction with their jobs as recyclers, reverse logistics service providers, seamstresses, designers and cooperatives in general. However, there is still a lack of jobs to reduce hunger and true social action.

It is concluded that of the ten I4.0Ts evaluated, cloud computing, cybersecurity systems, and additive manufacturing did not stimulate CE actions and performance.

Large-scale textile industries have indicated that security concerns, difficulty in rapid adaptation, and cost-related connectivity issues are the primary barriers to the adoption of cloud computing, thus not stimulating CE actions and sustainable performance. The first barrier pertains to security and compliance issues, as managers perceive cloud data storage to be inherently less secure than local data hosting. The second barrier is the challenge of swiftly transitioning and leveraging legacy technologies and systems, as cloud computing is not a plug-and-play solution, requiring a highly strategic and meticulous approach to implementation. It is noteworthy that technological changes have a short lifecycle, leading to hardware obsolescence, which would adversely affect circular economy actions. The third barrier is the lack of financial resources for investment in cloud computing, given that each company has likely already invested in both capital and operational expenses not only in the current IT infrastructure but also in IT professionals to manage it. Consequently, textile industry managers have recently invested in local technology and expect to recover their assets in the coming years before embarking on their journey to the cloud. Additionally, data centers supporting cloud computing consume significant amounts of energy, negatively impacting circular economy actions.

In this context, as previously mentioned, cloud computing was excluded from the measurement model due to the difficulties associated with adopting cybersecurity systems

to protect information. Large-scale textile industry managers understand that cloud computing must be implemented in conjunction with cybersecurity systems. They acknowledge the necessity of improving cyber control; without this control, data could be susceptible to hacking or leakage. As a result, they do not plan to adopt cybersecurity systems in the short term, primarily because the main barrier indicated by managers is the high cost of implementation. Furthermore, Information Security involves not only software and IT strategies but also a set of practices that everyone must adopt daily, highlighting the barrier related to the lack of stakeholder awareness and the scarcity of specialists in large-scale textile industries. Managers also mentioned that cybersecurity systems lead to rapid obsolescence in the supporting infrastructure (hardware), generating waste in a very short period, contrary to circular economy actions.

Additionally, in LTILB, additive manufacturing is relatively unknown and lacks an operational strategy for its feasibility in textile products. However, internationally, additive manufacturing in the industry is applied in the apparel chain, focusing on customization and personal style to contribute to CECF. The main barriers for LTILB to adopt additive manufacturing are related to: (i) the lack of an integrated digital process chain from design to sale in interface with the customer, applied in the apparel sector, aiming at customization in personal style; (ii) high investment costs, considering the hiring of specialists to design the digital apparel chain with additive manufacturing, as well as the technological infrastructure; and (iii) lack of qualified personnel to develop the IT architecture of the digital apparel chain, denoting limited knowledge about technologies and design for the application of additive manufacturing. In this context, there remains a significant journey for LTILB to adopt additive manufacturing in the apparel chain. Despite this, it represents an important strategy to reduce waste generation in the apparel industry, contributing to the circular economy.

The cyber physical system was the most applied technology with a focus on sensing and monitoring in real time to eliminate waste generation in the dyeing and manufacturing processes. Autonomous robots are being deployed in the processes of feeding machines, transporting materials, identifying, and segregating textile materials, and performing splicing, sewing, appliqué and whole piece making activities, as well as for material selection and suppliers of renewable materials and intelligent recycling.

The use of augmented reality optimizes the production schedule to generate less waste through virtual simulation before application in dyeing and apparel factories, as well as developing virtual molds of the product design, aiming at dematerialization and packaging reduction. The use of



big data analytics favors decision making in textile production, enabling the selection of suppliers to substitute productive resources for renewable ones and routing of reverse logistics for intelligent recycling. For the effectiveness of the big data analysis, it is important to apply simulation based on machine learning techniques to optimize production resources (materials and energy) of spinning, weaving, knitting, dyeing, and apparel processes. Consequently, using artificial intelligence to develop autonomous processes for classification and analysis of cotton quality, optimizing the generation of waste, which contributes to the reduction of residue and impact on the consumption of sustainable resources. It also suggests the use of IoT to assist the process of maintenance and real-time monitoring of operational processes that involve advanced technology equipment such as spinning and weaving mills.

While innovative, this study's limitation lies in its exclusive focus on LTILB; future research should explore small and medium-sized textile industries to elucidate the relationship between I4.0 T, CECP, and performance.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s13762-024-06010-y>.

Acknowledgements The authors are grateful to CNPq Conselho Nacional de Desenvolvimento Científico e Tecnológico (National Council for Scientific and Technological Development) – Research funding in Productivity – PQ-2 09/2023—in the field of Production and Transportation Engineering—Process: 305950/2023-1—for funding the study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

- Agarwal S, Singh AP (2022) Performance evaluation of textile wastewater treatment techniques using sustainability index: an integrated fuzzy approach of assessment. *J Clean Prod* 337:130384
- Alcayaga A, Wiener M, Hansen EG (2019) Towards a framework of smart-circular systems: an integrative literature review. *J Clean Prod* 221:622–634
- Angelis R, Feola R (2020) Circular business models in biological cycles: the case of an Italian spin-off. *J Clean Prod* 247:119603
- Aranda-Uson A, Portillo-Tarragona P, Scarpellini S, Llana-Macarulla F (2020) The progressive adoption of a circular economy by businesses for cleaner production: an approach from a regional study in Spain. *J Clean Prod* 247(20):119648
- Bai C, Dallasega P, Orzes G, Sarkis J (2020) Industry 4.0 technologies assessment: a sustainability perspective. *Int J Prod Econ* 229:107776
- Bloomfield M, Borstrock S (2018) Modeclix. The additively manufactured adaptable textile. *Mater Today Commun* 16:212–216
- Cassia AR, Costa I, da Silva VHC, de Oliveira Neto GC (2020) Systematic literature review for the development of a conceptual model on the relationship between knowledge sharing, information technology infrastructure and innovative capability. *Technol Anal Strategic Manag* 32(7):801–821
- Chen PY, Popovich PM (2002) Correlation: parametric and nonparametric measures, vol 139. SAGE Publications Inc., California
- Chen TL, Kim H, Pan SY, Tseng PC, Lin YP, Chiang PC (2020) Implementation of green chemistry principles in circular economy system towards sustainable development goals: Challenges and perspectives. *Sci Total Environ* 716:136998
- Chin WW (1998) The partial least squares approach to structural equation modeling. Lawrence Erlbawn, Mahwah
- Cohen J (1988) Statistical power analysis for the behavioral sciences, 2nd edn. Lawrence Erlbawn, Hillsdale
- Creswell, J.W (2014) Research design: qualitative, quantitative, and mixed methods approaches. In: research design qualitative quantitative and mixed methods approaches, 4th edn, SAGE Publications, Inc., Thousand Oaks
- Dancey, C.P., Reidy, J., (2007) Statistics without maths for psychology. Pearson Education
- De Oliveira Neto GC, Shibao FY, Filho MG, Chaves LEC (2015) Cleaner production: a study of the environmental and economic advantage in polymer recycling | Produção mais limpa: estudo da vantagem ambiental e econômica na reciclagem de polímeros. *Interciencia* 40(6):364–373
- de Oliveira Neto GC, Tucci HNP, Correia JMF, da Silva PC, da Silva D, Amorim M (2021) Stakeholders' influences on the adoption of cleaner production practices: a survey of the textile industry. *Sustain Prod Consum* 26:126–145
- de Oliveira Neto GC, da Conceição Silva A, Filho MG (2023) How can Industry 4.0 technologies and circular economy help companies and researchers collaborate and accelerate the transition to strong sustainability? A bibliometric review and a systematic literature review. *Int J Environ Sci Technol* 20(3):3483–3520
- de Paoli FM, de Oliveira Neto GC, Lucato WC (2013) Economic and environmental gains resulting from the utilization of the design for the environment (DfE) | Vantagens econômicas e ambientais resultantes da aplicação do projeto para o meio ambiente (DfE). *Espacios* 34(12):11
- Dillman DA, Smyth JD, Christian LM (2014) Internet, phone, mail, and mixed-mode surveys: the tailored design method. John Wiley & Sons, Hoboken
- Elkington J (1997) Cannibals with forks: the triple bottom line of 21st. Capstone Publishing Limited, Century Business. London
- Fahimnia B, Sarkis J, Gunasekaran A, Farahani R (2017) Decision models for sustainable supply chain design and management. *Ann Oper Res* 250(2):277–278
- Faria R, Lopes I, Pires IM, Marques G, Fernandes S, Garcia NM, Trajkovic V (2020) Circular economy for clothes using web and mobile technologies—a systematic review and a taxonomy proposal. *Information* 11(3):161
- Fatimah YA, Govindan K, Murniningsih R, Setiawan A (2020) Industry 4.0 based sustainable circular economy approach for smart waste management system to achieve sustainable development goals: a case study of Indonesia. *J Clean Prod* 269:122263
- Forza C (2002) Survey research in operation management: a process-based perspective. *Int J Oper Prod Manag* 22(2):152–194
- Fowler FJ (2014) Survey research methods. Sage Publications, Washington
- Franco MA (2017) Circular economy at the micro level: a dynamic view of incumbents' struggles and challenges in the textile industry. *J Clean Prod* 168:833–845



- Ghisellini P, Cialani C, Ulgiati SA (2016) review on circular economy: the expected transition to a balanced interplay of environmental and economic systems. *J Clean Prod* 114(15):11–32
- Gomes, M.G., da Silva, V.H.C., Rodrigues Pinto, L.F., Facchini, F., de Oliveira Neto, G.C. (2020). Economic, environmental and social gains of the implementation of artificial intelligence at dam operations toward industry 4.0 principles. *Sustainability* (Switzerland),12(9), 3604
- Groves RM, Fowler FJ Jr, Couper MP, Lepkowski JM, Singer E, Tourangeau R (2009) Survey methodology. John Wiley & Sons
- Hair JF, Sarstedt M, Pieper TM, Ringle CM (2012) The use of partial least squares structural equation modeling in strategic management research: a review of past practices and recommendations for future applications. *Long Range Plan* 45:320–340
- Hair JF, Hult GTM, Ringle CM, Sarstedt M (2016) A primer on partial least squares structural equation modeling (PLS-SEM). Sage, Los Angeles
- Hair JF, Hult GTM, Ringle CM, Sarstedt M, Thiele KO (2017a) Mirror, mirror on the wall: a comparative evaluation of composite-based structural equation modeling methods. *J Acad Market Sci* 45(5):616–632
- Hair JF, Hult GTM, Ringle CM, Sarstedt M (2017) A Primer on partial least squares structural equation modeling (PLS-SEM). Sage Publications, Washington
- Hair JF, Risher JJ, Sarstedt M, Ringle CM (2019) When to use and how to report the results of PLS-SEM. *Eur Bus Rev* 31(1):2–24
- Hens L, Block C, Cabello-Eras JJ, Sagastume-Gutiérrez A, García Lorenzo D, Chamorro C, Vandecasteele C (2018) On the evolution of cleaner production as a concept and a practice. *J Clean Prod* 172:3323–3333
- Henseler J, Ringle CM, Sarstedt M (2015) A new criterion for assessing discriminant validity in variance-based structural equation modeling. *J Acad Mark Sci* 43(1):115–135
- Jia F, Yin S, Chen L, Chen X (2020) The circular economy in the textile and apparel industry: a systematic literature review. *J Clean Prod* 259:120728
- Jin Y, Song X, Tang J, Dong X, Ji H (2021) Business model of garment enterprises: a scientometric review. *Text Res J* 91(13–14):1609–1626
- Kalmykovaa Y, Sadagopanb M, Rosadoc L (2018) Circular economy—from review of theories and practices to development of implementation tools. *Resour Conserv Recycl* 135:190–201
- Kline RB (2015) Principles and practice of structural equation modeling. Guilford Publications, New York
- Korhonen J, Honkasalo A, Seppälä J (2018) Circular economy: the concept and its limitations. *Ecol Econ* 143:37–46
- Kumar L, Kamil I, Ahmad M, Naqvi SA, Deitch MJ, Amjad AQ, Sassanelli C (2022) In-house resource efficiency improvements supplementing the end of pipe treatments in textile SMEs under a circular economy fashion. *Environ Sci* 10:1002319
- Larsson JKJ (2018) Digital innovation for sustainable apparel systems: experiences based on projects in textile value chain development. *Res J Text Appar* 22(4):370–389
- Luthra S, Mangla SK (2018) Evaluating challenges to industry 4.0 initiatives for supply chain sustainability in emerging economies. *Process Saf Environ Prot* 117:168–179
- Macarthur, D. E., Zumwinkel, K., Stuchtey, M. R. Growth within: a circular economy vision for a competitive europe. report of ellen macarthur foundation, (2015). Disponível em: https://www.ellenmacarthurfoundation.org/assets/downloads/publications/EllenMacArthurFoundation_Growth-Within_July15.pdf. Accessed on: July, 20, 2022
- McDowall W, Geng Y, Huang B, Barteková E, Bleischwitz R, Türkeli S, Doménech T (2017) Circular economy policies in China and Europe. *J Ind Ecol* 21(3):651–661
- Muredzi P (2015) Applying cleaner production and optimising heat process operations in dairy processing plants: a review. *Eur Sci J, Special Edit* 3:110–131
- Oliveira Neto GCD, Tucci HNP, Correia JMF, Da Silva PC, Da Silva VHC, Ganga GMD (2020) Assessing the implementation of cleaner production and company sizes: Survey in textile companies. *J Eng Fibers Fabr* 15:1–16
- Oliveira Neto GCD, Cesar da Silva P, Tucci HNP, Amorim M (2021) Reuse of water and materials as a cleaner production practice in the textile industry contributing to blue economy. *J Clean Prod* 305:127075
- Oliveira Neto GC, Correia JMF, Tucci HNP, Librantz AFH, Giannetti BF, Almeida CMVB (2022) Sustainable resilience degree assessment of the textile industrial by size: Incremental change in cleaner production practices considering circular economy. *J Clean Prod* 380:134633
- Periyasamy AP, Wiener J, Militky J (2017) Life-cycle assessment of denim. In: Muthu SS (ed) Sustainability in denim. Woodhead Publishing, Cambridge, UK, pp 83–110
- Pinto LF, Venturini GD, Digiesi S, Facchini F, Oliveira Neto GC (2020) Sustainability assessment in manufacturing under a strong sustainability perspective—an ecological neutrality initiative. *Sustainability* (Switzerland) 12(21):9232
- Rajput S, Singh SP (2019) Connecting circular economy and industry 4.0. *Int J Inf Manage* 49:98–113
- Rönkkö M, Evermann J (2013) A critical examination of common beliefs about partial least squares path modeling. *Organ Res Methods* 16(3):425–448
- Rönkkö M, McIntosh CN, Antonakis J, Edwards JR (2016) Partial least squares path modeling: time for some serious second thoughts. *J Oper Manag* 47:9–27
- Rüßmann, M.; Lorenz, M.; Gerbert, P.; Waldner, M.; Justus, J.; Engel, P.; Harnisch, M (2015) Industry 4.0: The Future of Productivity and Growth in Manufacturing Industries; Boston Consulting Group: Boston, MA, USA, 2015; pp. 1–14. Disponível em: https://image-src.bcg.com/Images/Industry_40_Future_of_Productivity_April_2015_tcm9-61694.pdf. Acesso em 20 de Julho de 2022
- Sarstedt M, Hair JF, Ringle CM, Thiele KO, Gudergan SP (2016) Estimation issues with PLS and CBSEM: where the bias lies! *J Business Res* 69:3998–4010
- Schumacker RE, Lomax RG (2015) A beginner's guide to structural equation modeling. Routledge
- Shayganmehr M, Kumar A, Garza-Reyes JA, Moktadir MA (2021) Industry 4.0 enablers for a cleaner production and circular economy within the context of business ethics: a study in a developing country. *J Clean Prod* 281:125280
- Shirvanimoghaddam K, Motamed B, Ramakrishna S, Naebe M (2020) Death by waste: Fashion and textile circular economy case. *Sci Total Environ* 718:137317
- Silva WDO, Morais DC (2022) Impacts and insights of circular business models' outsourcing decisions on textile and fashion waste management: a multi-criteria decision model for sorting circular strategies. *J Clean Prod* 370:133551
- Silva PC, Oliveira Neto GC, Correia JMF, Tucci HNP (2021) Evaluation of economic, environmental and operational performance of the adoption of cleaner production: survey in large textile industries. *J Clean Prod* 278:123855
- Sousa-Zomer TT, Magalhaes M, Zancul E, Campos LMS, Cauchick-Miguel PA (2018) Cleaner production as an antecedent for circular economy paradigm shift at the micro-level: Evidence from a home appliance manufacturer. *J Clean Prod* 185:740–748



- Tsai WH (2018) Green production planning and control for the textile industry by using mathematical programming and industry 4.0 techniques. *Energies* 11(8):2072
- Vandenberg RJ (2006) Introduction: statistical and methodological myths and urban legends: where, pray tell, did they get this idea? *Organ Res Methods* 9(2):194–201
- Westland JC (2010) Lower bounds on sample size in structural equation modeling. *Electron Commer Res Appl* 9(6):476–487

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

